

Human audit packet

Neuron model validation study: held-out evaluation

Naithik Somisetti · 25 faults · prepared 2026-09-22

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Date completed	2026-09-22
I worked alone and did not see the automated check (Yes / No)	No

1. What you are doing and why

We took six published computer models of single neurons and made small, deliberate edits ("faults") to them. Then we checked whether the usual test (re-running the one simulation shipped with each model, called the canonical test) notices the change, and whether a short set of extra stimuli does better.

The paper can only rely on these faults if a person has checked that they are what the software says they are. That is your job. You are checking three things for each fault: was the edit made as labelled, is the category it was put in believable, and is the evidence that it was detected believable.

Your answers are the independent check. Disagreeing with the software is a useful result, not a failure. If something looks wrong or you are not sure, say so.

2. Rules

- Work on your own. Do not compare answers with each other until you have both submitted.
- Do not ask Neel or any AI tool what the answer should be. Use only this packet. (A separate automated check exists; you will see it after you submit, so it cannot influence you.)
- "Unsure" is an allowed answer. Guessing "yes" is worse than an honest "unsure".
- Write a short note whenever you answer No or Unsure, saying what you saw.
- Expected time: about 1 to 2 hours. You do not need to finish in one sitting.

3. The ten-minute background

Neuron model. A set of equations describing how a nerve cell's voltage changes. Its ion channels (sodium, potassium, calcium and others) let current in or out. Each channel has settings such as how many channels there are (conductance), the voltage their current pushes toward (reversal potential), and how fast their gates open and close.

Stimulus (protocol). A pattern of current injected into the cell, like a step or a ramp. The cell responds by changing voltage and, if pushed hard enough, firing spikes (sharp upward blips, the tall lines in the graphs).

Features. Numbers measured from the voltage: how many spikes, when the first spike happens, the gaps between spikes, spike height, and so on.

Tolerance. How big a difference must be before it counts. It is set from the original model alone, so that tiny numerical noise never counts as a change.

h and h/2. The simulation is run twice, with a normal time step (h) and half that step (h/2). A difference only counts as a detection if it shows up at both. Some faults were checked a third time at h/4.

Two separate kinds of evidence. The assigned class is decided from the summary features only. Separately, the whole voltage trace is compared (spike count, spike timing, overall voltage difference). These are kept apart on purpose. So a fault can be class 4 ("no feature change") and still show whole-trace differences: that means the change was only visible in the raw trace. This is expected, not a contradiction.

The stimuli used:

Name in the tables	What it is
P00_canonical	the canonical test: the simulation shipped with the model
P02_weak_step	weak step: a small current step at half the firing threshold
P03_rheobase	rheobase search: finds the smallest current step that makes the cell fire
P04_step_2x	2x step: a current step at twice the firing threshold, 0.5 s
P05_long_step	long step: a current step at 1.5x the firing threshold, held for 2 s
P06_ramp	ramp: current rising steadily from 0 to 3x the firing threshold over 1 s
P07_hyperpolarizing_step	hyperpolarising step: a negative current step
P08_rebound	rebound: a negative step, then release
P09_short_pulse	short pulse: a 3 ms strong pulse

Categories (the "assigned class"):

Class	Meaning
1_structurally_invalid	the edited model file is no longer valid NeuroML
2_non_executable	the edited model would not run
3_numerically_unstable	the edited model ran but its numbers blew up
4_equivalent_within_tested_domain	no reproducible change in any summary feature (behaviour looks unchanged)
5_non_equivalent	behaviour changed, and the canonical test caught it
6_silent_under_canonical	behaviour changed, but the canonical test missed it (another protocol caught it)

4. How to answer the three questions

Question 1: Does the edit match its label?

Each fault page shows the exact lines of the model file before (–, red) and after (+, green) the edit. Check that the change is what the label says. For example, "multiply conductance by 0.5" should turn 0.0675 into 0.03375 in the condDensity value of one channel, and nothing else should change. Answer No if the numbers do not match the label, the wrong thing changed, or extra things changed.

Question 2: Is the assigned class plausible?

Compare the class with the feature evidence (the first evidence table). Remember that the class ignores whole-trace differences (section 3). If the class says the canonical test missed it, the canonical test (P00_canonical) should not appear among the detections at both h and h/2, but some other stimulus should. If the class says it would not run, there should be no measurements. If it says no change, there should be no detections at both step sizes.

Question 3: Is the detection plausible?

Look at the evidence tables and the graph. A detection means the difference is bigger than the tolerance, at h and at h/2. In the graph, black is the original model and orange dashed is the edited one: where they separate, the behaviour changed. For the key faults, the top graph (canonical test) should look nearly identical while the lower graph shows the difference. Answer N/A if nothing was detected.

5. How to submit

1. Save a copy of this document with your name in the file name, for example Audit_Packet_Samyak.docx.
2. Fill in the yellow box on the first page: your name, the date, and whether you worked alone.
3. Each fault's section ends with a yellow answer box. Type an X in one cell per row, and write a note whenever you answer No or Unsure.
4. Send your finished document back to Neel. Only then compare answers with each other.

6. Summary of the faults

#	Group	Model	Change	Assigned class
1	key fault (headline claim)	hay2011_soma	scale_conductance	6_silent_under_canonical
2	key fault (headline claim)	hay2011_soma	scale_conductance	4_equivalent_within_tested_domain
3	key fault (headline claim) + random sample	hay2011_soma	scale_gate_time_constant	6_silent_under_canonical
4	key fault (headline claim)	migliore2005_ca1_soma	scale_gate_time_constant	4_equivalent_within_tested_domain
5	key fault (headline claim)	bbp2015_soma	scale_gate_time_constant	6_silent_under_canonical
6	key fault (headline claim)	hay2011_soma	shift_reversal	4_equivalent_within_tested_domain
7	random sample	smith2013_singlecomp	duplicate_conductance	4_equivalent_within_tested_domain
8	random sample	migliore2005_ca1_soma	duplicate_conductance	5_non_equivalent
9	random sample	traub2005_testseg2	increase_dt	6_silent_under_canonical

#	Group	Model	Change	Assigned class
10	random sample	smith2013_singlecomp	increase_dt	6_silent_under_canonical
11	random sample	traub2005_testseg_all	scale_capacitance	5_non_equivalent
12	random sample	traub2005_testseg_all	scale_capacitance	5_non_equivalent
13	random sample	migliore2005_ca1_soma	scale_conductance	5_non_equivalent
14	random sample	bbp2015_soma	scale_conductance	6_silent_under_canonical
15	random sample	migliore2005_ca1_soma	scale_conductance	6_silent_under_canonical
16	random sample	traub2005_testseg2	scale_conductance	5_non_equivalent
17	random sample	bbp2015_soma	scale_gate_slope	5_non_equivalent
18	random sample	smith2013_singlecomp	scale_gate_time_constant	6_silent_under_canonical
19	random sample	traub2005_testseg_all	shift_forward_rate_midpoint	5_non_equivalent
20	random sample	traub2005_testseg2	shift_gate_midpoint	5_non_equivalent
21	random sample	migliore2005_ca1_soma	shift_gate_midpoint	5_non_equivalent
22	random sample	traub2005_testseg_all	shift_reversal	5_non_equivalent
23	random sample	bbp2015_soma	solver_config	4_equivalent_within_tested_domain
24	random sample	hay2011_soma	wrong_channel	2_non_executable
25	random sample	migliore2005_ca1_soma	wrong_compatible_component	3_numerically_unstable

Fault 1 of 25

Fault ID	m-scale_conductance-211745202a
Group	key fault (headline claim)
Model	Hay et al. 2011 layer-5 pyramidal cell (soma only) (hay2011_soma)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 1.1
Severity	mild
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P05_long_step

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/Soma_AllCML.cell.nml

```
@@ -84,5 +84,5 @@
<!--Reversal potential for ca in Ca_HVA will be calculated by Nernst equation from internal & external calcium-->

- <channelDensityNernst condDensity="0.992 mS_per_cm2" id="Ca_HVA_all" ionChannel="Ca_HVA" ion="ca"/>
+ <channelDensityNernst condDensity="1.0912 mS_per_cm2" id="Ca_HVA_all" ionChannel="Ca_HVA" ion="ca"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

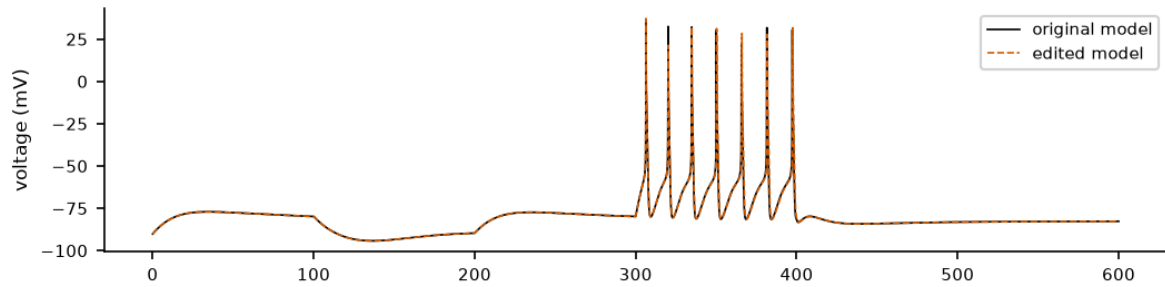
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P05_long_step	last_isi	143	148.3	5.33	3.03
h	P05_long_step	spike_count	14	13	1	0.5
h/2	P05_long_step	last_isi	144	149.3	5.36	3.03
h/2	P05_long_step	spike_count	14	13	1	0.5

Whole-voltage-trace differences that exceeded tolerance:

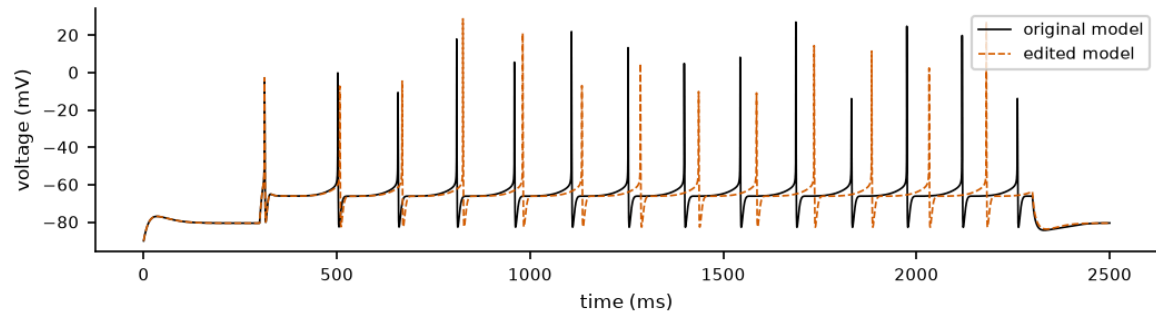
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P05_long_step	spike count	14	13	1	0.5
h	P06_ramp	spike timing (largest shift, ms)	–	–	7.177	3.955
h/2	P05_long_step	spike count	14	13	1	0.5
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	7.155	3.955
h/4	P05_long_step	spike count	14	13	1	0.5
h/4	P06_ramp	spike timing (largest shift, ms)	–	–	7.146	3.955

Fault 1: m-scale_conductance-211745202a on hay2011_soma

the canonical test: the simulation shipped with the model



long step: a current step at 1.5x the firing threshold, held for 2 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 1 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	—
Notes (required for any No or Unsure):				

Fault 2 of 25

Fault ID	m-scale_conductance-75492a609a
Group	key fault (headline claim)
Model	Hay et al. 2011 layer-5 pyramidal cell (soma only) (hay2011_soma)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 0.5
Severity	strong
Run status	ok
Assigned class	4_equivalent_within_tested_domain: no reproducible change in any summary feature (behaviour looks unchanged)
Detected (features) by	nothing

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/Soma_AllCML.cell.nml

@@ -65,5 +65,5 @@

<channelDensity condDensity="0.2 mS_per_cm2" id="Ih_all" ionChannel="Ih" ion="hcn" erev="-45.0 mV"/>

- <channelDensity condDensity="0.0675 mS_per_cm2" id="Im_all" ionChannel="Im" ion="k" erev="-85.0 mV"/>

+ <channelDensity condDensity="0.03375 mS_per_cm2" id="Im_all" ionChannel="Im" ion="k" erev="-85.0 mV"/>

<channelDensity condDensity="2.23 mS_per_cm2" id="K_Pst_all" ionChannel="K_Pst" ion="k" erev="-85.0 mV"/>

Evidence (Questions 2 and 3)

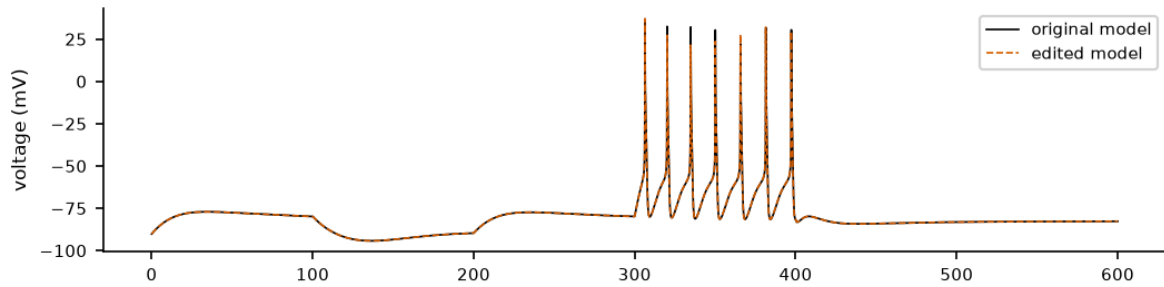
No feature differences exceeded tolerance.

Whole-voltage-trace differences that exceeded tolerance:

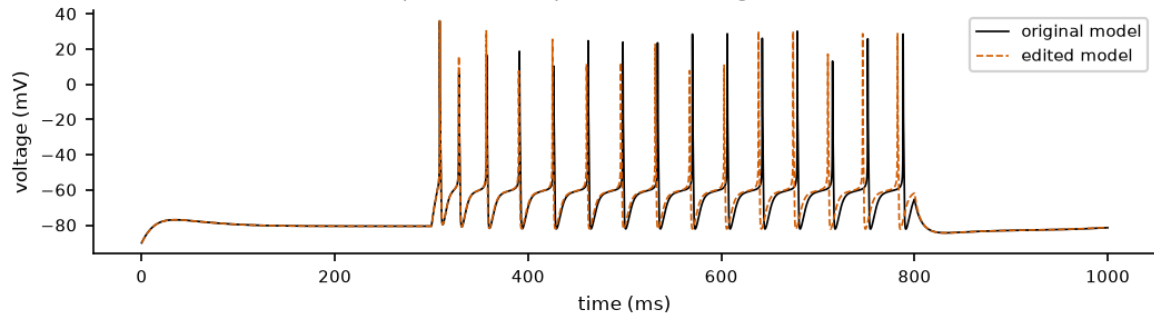
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike timing (largest shift, ms)	–	–	5.773	3.496
h/2	P04_step_2x	spike timing (largest shift, ms)	–	–	5.768	3.496
h/4	P04_step_2x	spike timing (largest shift, ms)	–	–	5.765	3.496

Fault 2: m-scale_conductance-75492a609a on hay2011_soma

the canonical test: the simulation shipped with the model



2x step: a current step at twice the firing threshold, 0.5 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 2 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?		X		
Confidence: type X under one	High X	Medium	Low	—
Notes (required for any No or Unsure): 0.0675 x 0.5 = 0.03375 exactly. Header states “Detected (features) by: nothing.” Class 4 was satisfied.				

Fault 3 of 25

Fault ID	m-scale_gate_time_constant-02eb4e0b90
Group	key fault (headline claim) + random sample
Model	Hay et al. 2011 layer-5 pyramidal cell (soma only) (hay2011_soma)
Label (what the edit claims to be)	scale_gate_time_constant: make one channel gate open and close faster or slower (scale its time constant); factor 1.25
Severity	mild
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P05_long_step

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/Ca_HVA.channel.nml

@@ -36,6 +36,6 @@

```
<gate id="m" type="gateHHrates" instances="2">
- <forwardRate type="HHExpLinearRate" rate="0.209per_ms" scale="3.8mV" midpoint="-27mV"/>
- <reverseRate type="HHExpRate" rate="0.94per_ms" scale="-17mV" midpoint="-75mV"/>
+ <forwardRate type="HHExpLinearRate" rate="0.26125per_ms" scale="3.8mV" midpoint="-27mV"/>
+ <reverseRate type="HHExpRate" rate="1.175per_ms" scale="-17mV" midpoint="-75mV"/>
</gate>
```

Evidence (Questions 2 and 3)

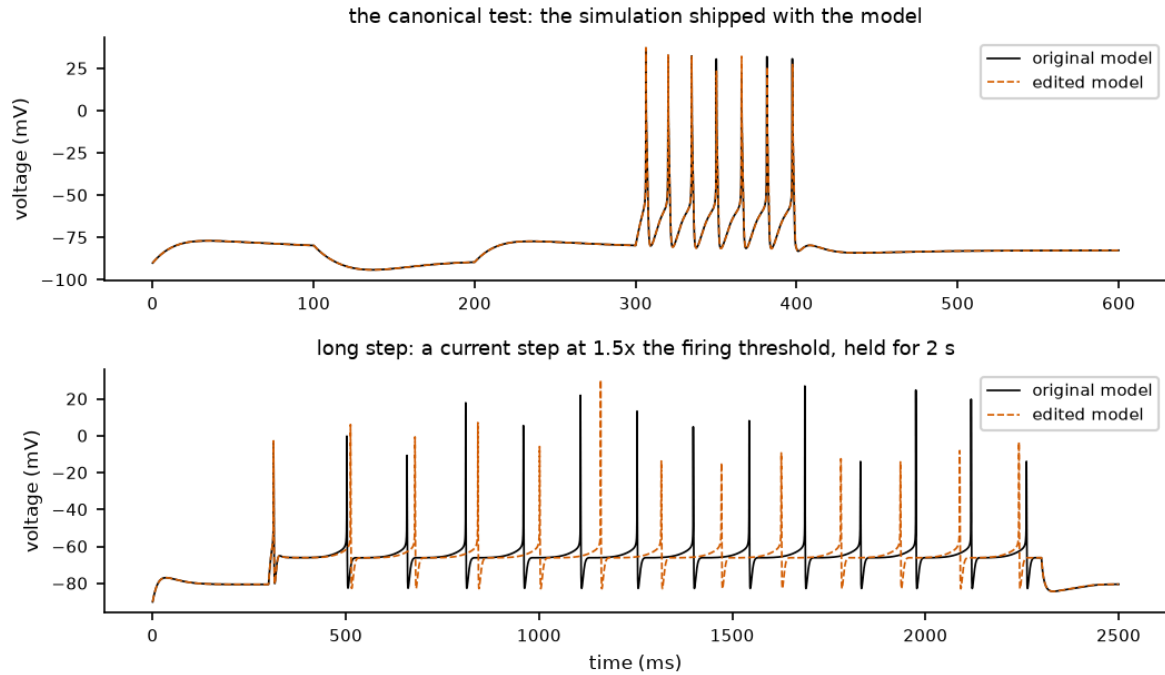
Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P05_long_step	last_isi	143	153.4	10.44	3.03
h	P05_long_step	spike_count	14	13	1	0.5
h/2	P05_long_step	last_isi	144	154.4	10.38	3.03
h/2	P05_long_step	spike_count	14	13	1	0.5

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P05_long_step	spike count	14	13	1	0.5
h	P06_ramp	spike timing (largest shift, ms)	–	–	13.8	3.955
h/2	P05_long_step	spike count	14	13	1	0.5
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	13.59	3.955
h/4	P05_long_step	spike count	14	13	1	0.5
h/4	P06_ramp	spike timing (largest shift, ms)	–	–	13.5	3.955

Fault 3: m-scale_gate_time_constant-02eb4e0b90 on hay2011_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 3 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	—
Notes (required for any No or Unsure):				

Fault 4 of 25

Fault ID	m-scale_gate_time_constant-31e82d8467
Group	key fault (headline claim)
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	scale_gate_time_constant: make one channel gate open and close faster or slower (scale its time constant); factor 0.8
Severity	mild
Run status	ok
Assigned class	4_equivalent_within_tested_domain: no reproducible change in any summary feature (behaviour looks unchanged)
Detected (features) by	nothing

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/kad.channel.nml

@@ -40,5 +40,5 @@

```
<gate id="1" type="gateHHTauInf" instances="1">
- <q10Settings type="q10Fixed" fixedQ10="1"/>
+ <q10Settings type="q10Fixed" fixedQ10="0.8"/>
<timeCourse type="kad_l_tau_tau"/>
<steadyState type="kad_l_inf_inf"/>
```

Evidence (Questions 2 and 3)

No feature differences exceeded tolerance.

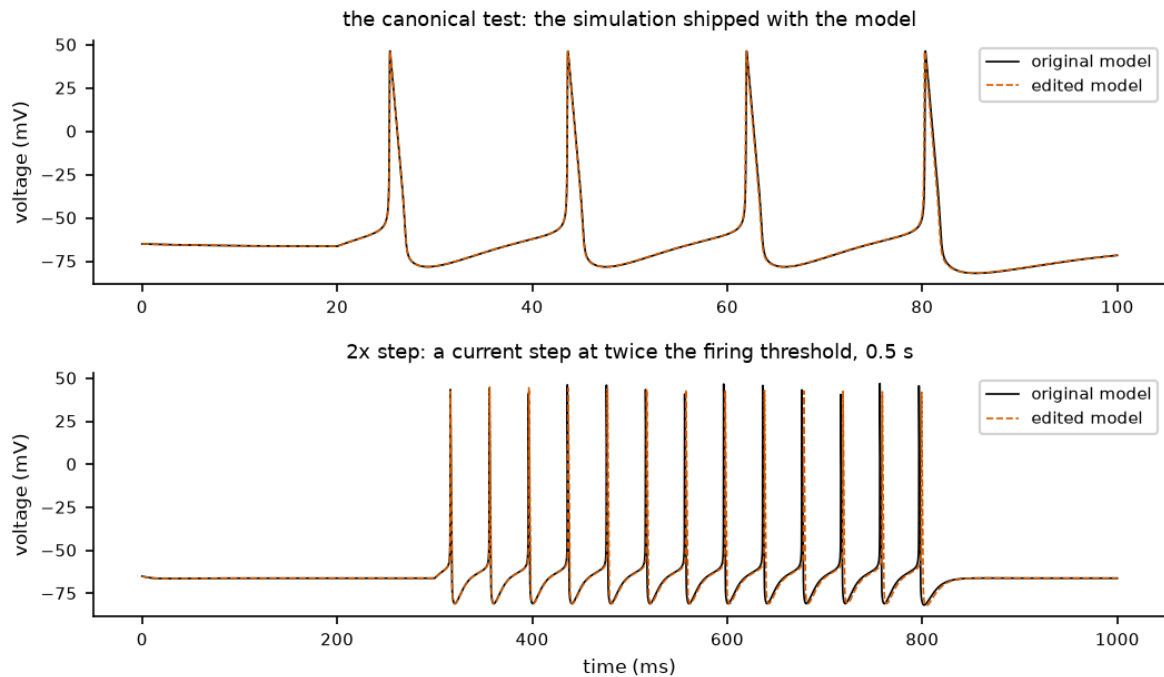
Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P04_step_2x	overall voltage difference (RMS, mV)	0	12.83	12.83	8.749
h	P04_step_2x	spike timing (largest shift, ms)	–	–	2.581	0.5184
h	P05_long_step	spike timing (largest shift, ms)	–	–	13.69	4.506
h	P06_ramp	overall voltage difference (RMS, mV)	0	13.15	13.15	6.268
h	P06_ramp	spike timing (largest shift, ms)	–	–	2.402	0.5781
h/2	P04_step_2x	overall voltage difference (RMS, mV)	0	12.8	12.8	8.749
h/2	P04_step_2x	spike timing (largest shift, ms)	–	–	2.581	0.5184
h/2	P05_long_step	spike timing (largest shift, ms)	–	–	13.72	4.506
h/2	P06_ramp	overall voltage difference (RMS, mV)	0	13.11	13.11	6.268
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	2.398	0.5781
h/4	P04_step_2x	overall voltage difference (RMS, mV)	0	12.78	12.78	8.749
h/4	P04_step_2x	spike timing (largest shift, ms)	–	–	2.582	0.5184

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h/4	P05_long_step	spike timing (largest shift, ms)	–	–	13.74	4.506
h/4	P06_ramp	overall voltage difference (RMS, mV)	0	13.09	13.09	6.268

(1 more rows not shown)

Fault 4: m-scale_gate_time_constant-31e82d8467 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 4 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?			X	–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?		X		
Confidence: type X under one	High	Medium X	Low	–

Notes (required for any No or Unsure):

The edit sets fixed Q10 from 1 to 0.8. This is an implementation of scale time constant by 0.8, but Q10 scaling is temperature dependent, so I am not fully sure that the resulting T scale factor is exactly 0.8 from the diff alone.

Class 4 is satisfied, and the large whole-trace-only differences are explicitly stated by Section 3 as trace only detections and not a contradiction. Header confirms that Detected (features) by: nothing. Q3 is N/A

Fault 5 of 25

Fault ID	m-scale_gate_time_constant-de97b02e71
Group	key fault (headline claim)
Model	Blue Brain Project 2015 cortical cell (soma only) (bbp2015_soma)
Label (what the edit claims to be)	scale_gate_time_constant: make one channel gate open and close faster or slower (scale its time constant); factor 1.25
Severity	mild
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P05_long_step

The exact change (Question 1)

File: NMC/NeuroML2/Ca_HVA.channel.nml

@@ -36,6 +36,6 @@

```
<gate id="m" type="gateHHrates" instances="2">
- <forwardRate type="HHExpLinearRate" rate="0.209per_ms" scale="3.8mV" midpoint="-27mV"/>
- <reverseRate type="HHExpRate" rate="0.94per_ms" scale="-17mV" midpoint="-75mV"/>
+ <forwardRate type="HHExpLinearRate" rate="0.26125per_ms" scale="3.8mV" midpoint="-27mV"/>
+ <reverseRate type="HHExpRate" rate="1.175per_ms" scale="-17mV" midpoint="-75mV"/>
</gate>
```

Evidence (Questions 2 and 3)

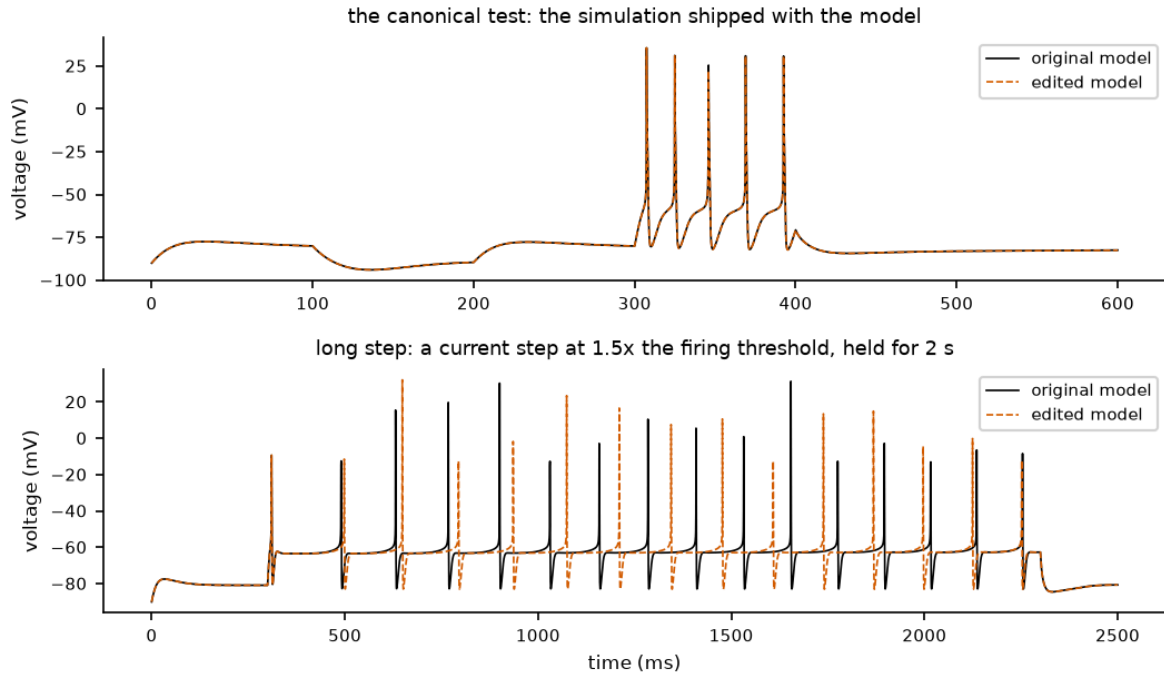
Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P05_long_step	last_isi	118.7	127.3	8.59	2.373
h	P05_long_step	spike_count	16	15	1	0.5
h/2	P05_long_step	last_isi	118.9	127.4	8.52	2.373
h/2	P05_long_step	spike_count	16	15	1	0.5

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P05_long_step	spike count	16	15	1	0.5
h	P06_ramp	spike timing (largest shift, ms)	–	–	8.159	2.334
h/2	P05_long_step	spike count	16	15	1	0.5
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	8.087	2.334
h/4	P05_long_step	spike count	16	15	1	0.5
h/4	P06_ramp	spike timing (largest shift, ms)	–	–	8.052	2.334

Fault 5: m-scale_gate_time_constant-de97b02e71 on bbp2015_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 5 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	—
Notes (required for any No or Unsure):				

Fault 6 of 25

Fault ID	m-shift_reversal-e5f14f8fb3
Group	key fault (headline claim)
Model	Hay et al. 2011 layer-5 pyramidal cell (soma only) (hay2011_soma)
Label (what the edit claims to be)	shift_reversal: shift one channel's reversal potential (the voltage its current pushes toward) by a few mV
Severity	strong
Run status	ok
Assigned class	4_equivalent_within_tested_domain: no reproducible change in any summary feature (behaviour looks unchanged)
Detected (features) by	nothing

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/Soma_AllCML.cell.nml

@@ -65,5 +65,5 @@

<channelDensity condDensity="0.2 mS_per_cm2" id="Ih_all" ionChannel="Ih" ion="hcn" erev="-45.0 mV"/>

- <channelDensity condDensity="0.0675 mS_per_cm2" id="Im_all" ionChannel="Im" ion="k" erev="-85.0 mV"/>

+ <channelDensity condDensity="0.0675 mS_per_cm2" id="Im_all" ionChannel="Im" ion="k" erev="-95.0 mV"/>

<channelDensity condDensity="2.23 mS_per_cm2" id="K_Pst_all" ionChannel="K_Pst" ion="k" erev="-85.0 mV"/>

Evidence (Questions 2 and 3)

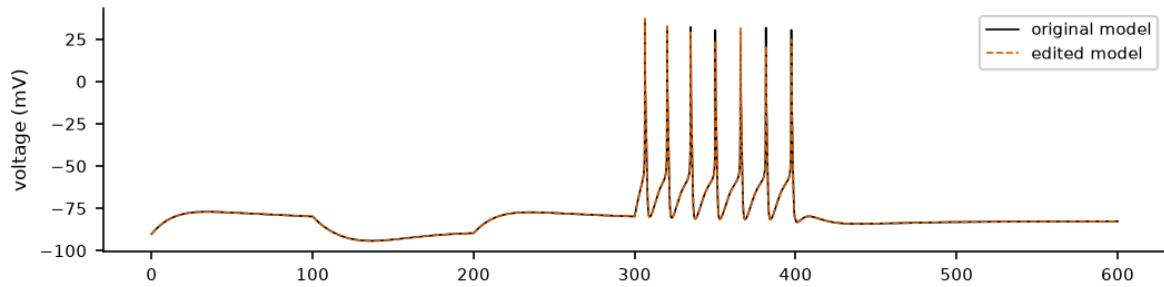
No feature differences exceeded tolerance.

Whole-voltage-trace differences that exceeded tolerance:

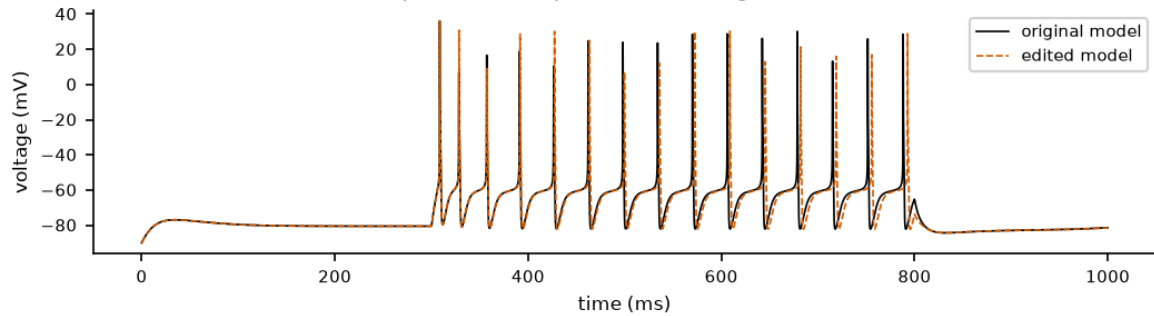
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike timing (largest shift, ms)	–	–	4.622	3.496
h/2	P04_step_2x	spike timing (largest shift, ms)	–	–	4.635	3.496
h/4	P04_step_2x	spike timing (largest shift, ms)	–	–	4.642	3.496

Fault 6: m-shift_reversal-e5f14f8fb3 on hay2011_soma

the canonical test: the simulation shipped with the model



2x step: a current step at twice the firing threshold, 0.5 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 6 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?			X	–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?		X		
Confidence: type X under one	High	Medium	Low	–
Notes (required for any No or Unsure): Erev -85 to 95mV is a 10mV shift. The label says by a few mV, which reads as a stretch for 10mV. However, Class 4 rule itself is satisfied. Header confirms that Detected (features) by: nothing. Q3 is N/A				

Fault 7 of 25

Fault ID	m-duplicate_conductance-49ae908693
Group	random sample
Model	Smith et al. 2013 layer-2/3 pyramidal cell (one compartment) (smith2013_singlecomp)
Label (what the edit claims to be)	duplicate_conductance: add a second copy of one channel (roughly doubles it)
Severity	not_applicable
Run status	ok
Assigned class	4_equivalent_within_tested_domain: no reproducible change in any summary feature (behaviour looks unchanged)
Detected (features) by	nothing

The exact change (Question 1)

File: NeuroML2/singleCompAllChans.cell.nml

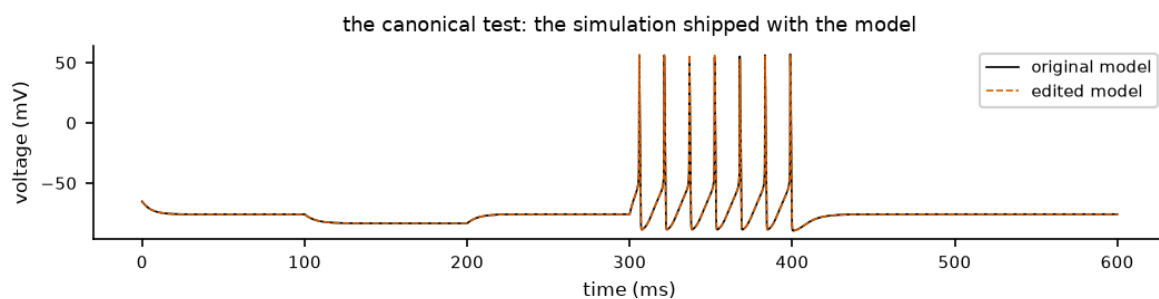
@@ -47,4 +47,5 @@

```
<channelDensity condDensity="2.2e-4 S_per_cm2" id="km_all" ionChannel="km" ion="k" erev="-90 mV"/>
<channelDensity condDensity="0.5e-4 S_per_cm2" id="ca_all" ionChannel="ca" ion="ca" erev="140.0 mV"/>
+ <channelDensity condDensity="0.5e-4 S_per_cm2" id="ca_all_dup" ionChannel="ca" ion="ca" erev="140.0 mV"/>
<channelDensity condDensity="3.0e-4 S_per_cm2" id="kca_all" ionChannel="kca" ion="k" erev="-90 mV"/>
<channelDensity condDensity="0.0003 S_per_cm2" id="it_all" ionChannel="it" ion="ca" erev="140.0 mV"/>
```

Evidence (Questions 2 and 3)

No feature differences exceeded tolerance.

Fault 7: m-duplicate_conductance-49ae908693 on smith2013_singlecomp



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 7 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?		X		

Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure): Clean single line duplicate added. Zero detections anywhere. unambiguous Class 4 and unambiguous N/A.				

Fault 8 of 25

Fault ID	m-duplicate_conductance-f1055b2803
Group	random sample
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	duplicate_conductance: add a second copy of one channel (roughly doubles it)
Severity	not_applicable
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P03_rheobase, P04_step_2x, P05_long_step, P06_ramp, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/SomaOnly_allCml.cell.nml

@@ -77,4 +77,6 @@

```
<channelDensity condDensity="10.0 mS_per_cm2" id="kdr_all" ionChannel="kdr" ion="k" erev="-90.0 mV"/>
```

```
+ <channelDensity condDensity="10.0 mS_per_cm2" id="kdr_all_dup" ionChannel="kdr" ion="k" erev="-90.0 mV"/>
```

```
+
```

```
<channelDensity condDensity="25.0 mS_per_cm2" id="na3_all" ionChannel="na3" ion="na" erev="50.0 mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	ahp_depth	-11.85	-16.48	4.627	0.5927
h	P00_canonical	last_isi	18.33	19.41	1.08	0.5
h	P03_rheobase	rheobase	5.806e-4	6.830e-4	1.025e-4	6.830e-5
h	P04_step_2x	ahp_depth	-14.78	-18.77	3.991	0.7388
h	P04_step_2x	first_spike_latency	16.07	17.58	1.51	0.5
h	P04_step_2x	last_isi	40.06	41.4	1.34	0.8012
h	P04_step_2x	spike_count	13	12	1	0.5
h	P05_long_step	ahp_depth	-15.07	-18.99	3.914	0.7537
h	P05_long_step	first_spike_latency	23.15	26.93	3.78	0.5
h	P05_long_step	last_isi	51.51	54.38	2.87	1.03
h	P05_long_step	spike_count	39	37	2	0.5
h	P06_ramp	first_spike_latency	396.2	450.3	54.06	7.924
h	P06_ramp	spike_count	16	15	1	0.5
h	P09_short_pulse	ahp_depth	-16	-19.59	3.588	0.8001
h/2	P00_canonical	ahp_depth	-11.8	-16.45	4.642	0.5927
h/2	P00_canonical	last_isi	18.32	19.4	1.08	0.5

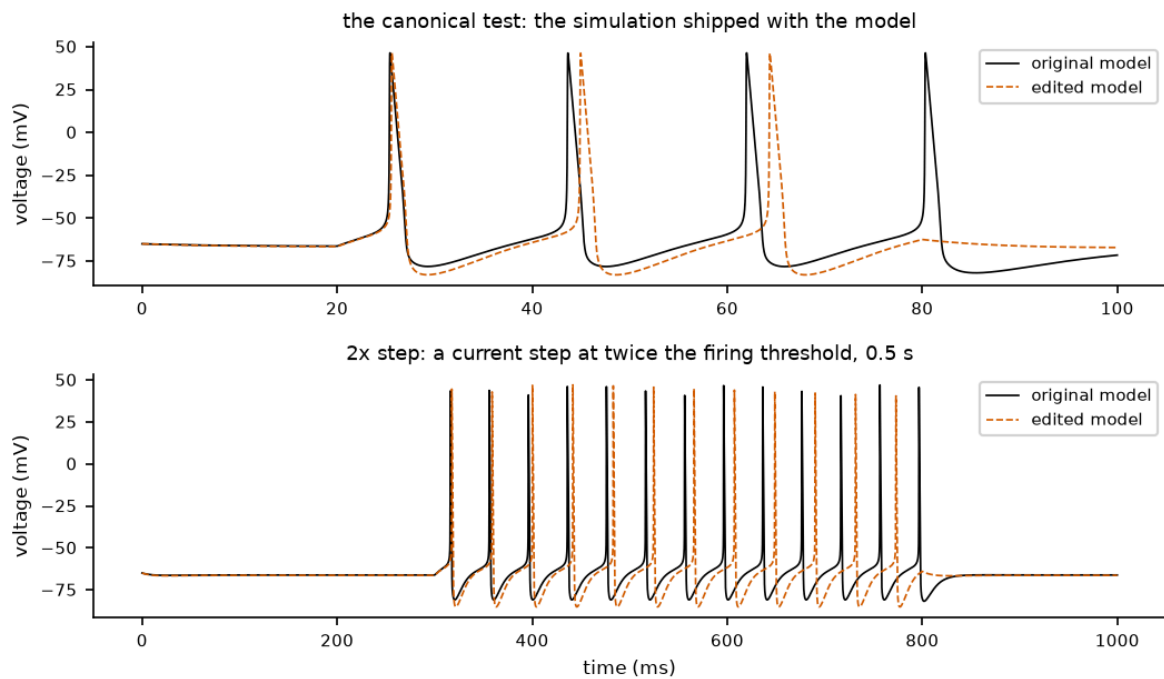
(12 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	21.21	21.21	4.056
h	P00_canonical	spike count	4	3	1	0.5
h	P04_step_2x	overall voltage difference (RMS, mV)	0	16.81	16.81	8.749
h	P04_step_2x	spike count	13	12	1	0.5
h	P05_long_step	spike count	39	37	2	0.5
h	P06_ramp	overall voltage difference (RMS, mV)	0	14.66	14.66	6.268
h	P06_ramp	spike count	16	15	1	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	1.587	1.587	1.226
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	21.19	21.19	4.056
h/2	P00_canonical	spike count	4	3	1	0.5
h/2	P04_step_2x	overall voltage difference (RMS, mV)	0	16.75	16.75	8.749
h/2	P04_step_2x	spike count	13	12	1	0.5
h/2	P05_long_step	spike count	39	37	2	0.5
h/2	P06_ramp	overall voltage difference (RMS, mV)	0	14.61	14.61	6.268

(2 more rows not shown)

Fault 8: m-duplicate_conductance-f1055b2803 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 8 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure):				

Fault 9 of 25

Fault ID	m-increase_dt-1086ebecdc
Group	random sample
Model	Traub et al. 2005 test cell 2 (sodium/potassium channels) (traub2005_testseg2)
Label (what the edit claims to be)	increase_dt: make the simulation time step larger (a numerical stress test, not a model fault); factor 4
Severity	strong
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P04_step_2x, P05_long_step, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/LEMS_SomaTest.xml

```
@@ -1,4 +1,3 @@
- <Lems xmlns="http://www.neuroml.org/lems/0.7.2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
-   xsi:schemaLocation="http://www.neuroml.org/lems/0.7.2
https://raw.githubusercontent.com/LEMS/LEMS/development/Schemas/LEMS/LEMS_v0.7.2.xsd">
+ <Lems xmlns="http://www.neuroml.org/lems/0.7.2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
+   xsi:schemaLocation="http://www.neuroml.org/lems/0.7.2
https://raw.githubusercontent.com/LEMS/LEMS/development/Schemas

<Target component="sim1" reportFile="simulator.props" timesFile="time.dat"/>
@@ -38,5 +37,5 @@
<!--Note: this could be: Simulation id="sim1" ... , but Component type="Simulation" ...
is used to allow validation of this file according to the LEMS schema specified above...-->
- <Component type="Simulation" id="sim1" length="100.0ms" step="0.001ms" target="network_Thalamocortical">
+ <Component type="Simulation" id="sim1" length="100.0ms" step="0.004ms" target="network_Thalamocortical">

<Display id="CG_tester_v" title="Thalamocortical: TestNML2, CG_tester, VOLTAGE" xmin="-0.010000001"
xmax="0.11000001" ymin="-0.104" ymax="0.064" timeScale="1s">
```

Evidence (Questions 2 and 3)

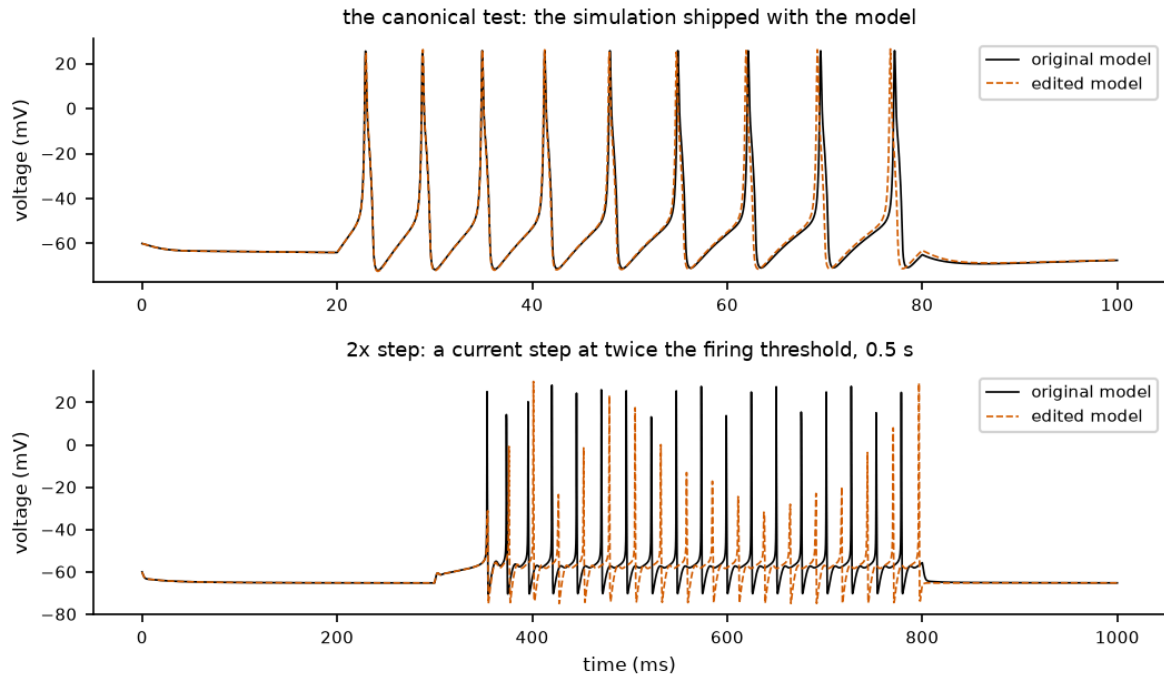
Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P04_step_2x	ahp_depth	-5.433	-9.857	4.424	1.145
h	P04_step_2x	ap_amplitude	76.68	79.82	3.13	1.896
h	P05_long_step	ahp_depth	-5.301	-9.83	4.529	1.153
h	P05_long_step	ap_amplitude	77.27	80.35	3.078	1.983
h	P09_short_pulse	ahp_depth	-8.488	-11.64	3.149	1.087
h	P09_short_pulse	ap_amplitude	90.84	—	—	1.817
h/2	P04_step_2x	ahp_depth	-5.052	-6.419	1.367	1.145
h/2	P04_step_2x	spike_count	18	19	1	0.5
h/2	P05_long_step	ahp_depth	-4.917	-6.305	1.388	1.153
h/2	P09_short_pulse	ahp_depth	-8.126	-9.304	1.178	1.087

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	7.838	7.838	4.39
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	0.7114	0.7114	0.5
h/2	P04_step_2x	spike count	18	19	1	0.5

Fault 9: m-increase_dt-1086ebecdc on traub2005_testseg2



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 9 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?			X	—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High	Medium X	Low	—
Notes (required for any No or Unsure):				

Step 0.001 to 0.004 matches factor 4. I am not treating the identical looking header line diff noise as a real second edit, but I can't fully rule it out. Class 6 is satisfied.

Fault 10 of 25

Fault ID	m-increase_dt-9ff98cb7be
Group	random sample
Model	Smith et al. 2013 layer-2/3 pyramidal cell (one compartment) (smith2013_singlecomp)
Label (what the edit claims to be)	increase_dt: make the simulation time step larger (a numerical stress test, not a model fault); factor 4
Severity	strong
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P09_short_pulse

The exact change (Question 1)

File: NeuroML2/LEMS_singleCompAllChans.xml

```
@@ -22,14 +22,14 @@
<!--Note: this could be: Simulation id="sim1" ... , but Component type="Simulation" ...
is used to allow validation of this file according to the LEMS schema specified above.-->
- <Component type="Simulation" id="sim1" length="600ms" step="0.001 ms" target="net">
+ <Component type="Simulation" id="sim1" length="600ms" step="0.004 ms" target="net">

- <Display id="" title="v" xmin="0" xmax="600" ymin="-100.0" ymax="100" timeScale="1ms">
+ <Display id="" title="v" xmin="0" xmax="600" ymin="-100.0" ymax="100" timeScale="1ms">
<Line id="VOLTAGE" timeScale="1ms" quantity="pop0/0/cell/v" scale="1 mV" color="#000000"/>
</Display>
<Display id="d2" title="Ca current density (uA_per_cm2)" timeScale="1ms" xmin="0" xmax="600" ymin="0" ymax="3">
- <Line id="Ca iDensity" quantity="pop0/0/cell/biophys/membraneProperties/ca_all/iDensity" scale="1uA_per_cm2"
color="#00ff00" timeScale="1ms"/>
+ <Line id="Ca iDensity" quantity="pop0/0/cell/biophys/membraneProperties/ca_all/iDensity" scale="1uA_per_cm2"
color="#00ff00" timeScale="1ms"/>
</Display>
<Display id="d3" title="Internal Ca concentration (mM)" timeScale="1ms" xmin="0" xmax="600" ymin="0" ymax="30">
- <Line id="caConc" quantity="pop0/0/cell/caConc" scale="1e-6 mM" color="#ff0000" timeScale="1ms"/>
+ <Line id="caConc" quantity="pop0/0/cell/caConc" scale="1e-6 mM" color="#ff0000" timeScale="1ms"/>
</Display>
<OutputFile id="outf" fileName="jlems_sccct.dat">
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P04_step_2x	ahp_depth	-12.95	134.5	147.4	0.6473
h	P09_short_pulse	ahp_depth	-13.72	123.2	137	0.6861
h	P09_short_pulse	ap_amplitude	134.2	–	–	2.683
h/2	P09_short_pulse	ap_amplitude	133.8	–	–	2.683

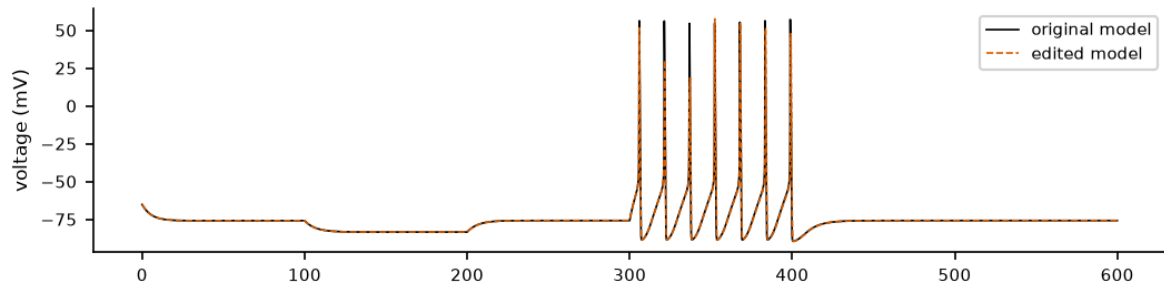
Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	4.237	4.237	3.009

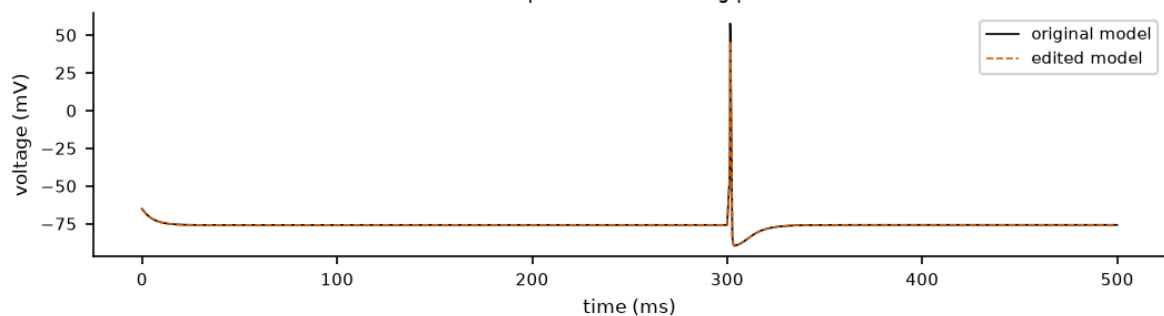
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike timing (largest shift, ms)	–	–	5.715	3.078
h	P06_ramp	spike timing (largest shift, ms)	–	–	5.6	3.018
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	0.8654	0.8654	0.7748

Fault 10: m-increase_dt-9ff98cb7be on smith2013_singlecomp

the canonical test: the simulation shipped with the model



short pulse: a 3 ms strong pulse



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 10 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?		X		–
Q3 Detection plausible?		X		
Confidence: type X under one	High	Medium X	Low	–

Notes (required for any No or Unsure):

0.001 to 0.004ms matches factor 4. Same header line diff noise caveat at fault 9. Class 6 is satisfied. P00_canonical absent from feature table; P09_short_pulse detected at both h,h2.

Fault 11 of 25

Fault ID	m-scale_capacitance-8d53b9976a
Group	random sample
Model	Traub et al. 2005 test cell with almost all channels (traub2005_testseg_all)
Label (what the edit claims to be)	scale_capacitance: multiply the membrane capacitance by a factor; factor 0.8
Severity	mild
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P04_step_2x, P05_long_step, P06_ramp, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/TestSeg_all.cell.nml

```
@@ -136,5 +136,5 @@
<spikeThresh value="0 mV"/>

- <specificCapacitance value="1.0 uF_per_cm2"/>
+ <specificCapacitance value="0.8 uF_per_cm2"/>

<initMembPotential value="-60.0 mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

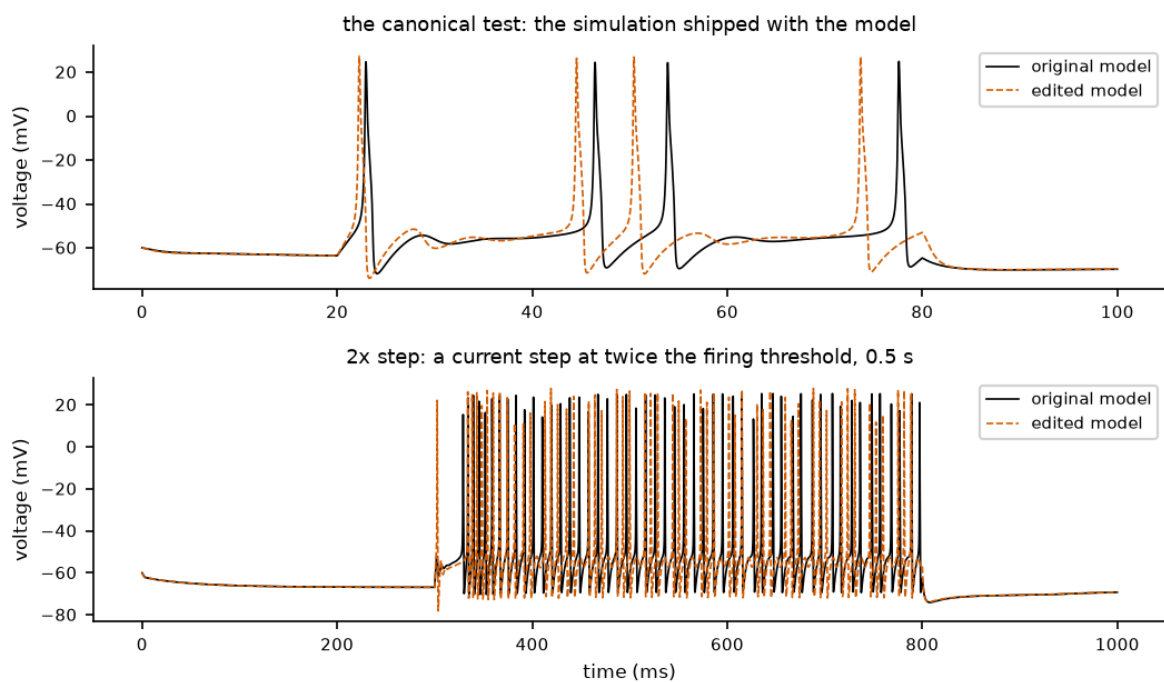
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	adaptation_index	0.5217	0.5962	0.07448	0.05217
h	P00_canonical	ahp_depth	-8.248	-10.33	2.078	0.5
h	P00_canonical	ap_amplitude	74.71	80.05	5.339	1.494
h	P00_canonical	first_spike_latency	2.95	2.27	0.68	0.5
h	P04_step_2x	ahp_depth	-3.064	-11.19	8.125	1.146
h	P04_step_2x	first_spike_latency	29.1	2.61	26.49	0.582
h	P05_long_step	ahp_depth	-2.799	-5.095	2.296	1.173
h	P05_long_step	ap_amplitude	72.99	75.78	2.787	1.46
h	P05_long_step	first_spike_latency	42.89	41.71	1.18	0.8578
h	P05_long_step	last_isi	24.47	22.67	1.8	1.11
h	P05_long_step	spike_count	83	90	7	3
h	P06_ramp	spike_count	70	78	8	3
h	P09_short_pulse	ahp_depth	-2.698	-4.395	1.697	1.327
h	P09_short_pulse	ap_amplitude	96.16	99.1	2.943	1.923
h/2	P00_canonical	adaptation_index	0.5215	0.596	0.07453	0.05217
h/2	P00_canonical	ahp_depth	-8.214	-10.29	2.074	0.5

(12 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	14.87	14.87	1.009
h	P00_canonical	spike timing (largest shift, ms)	–	–	3.909	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	2.463	2.463	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	14.87	14.87	1.009
h/2	P00_canonical	spike timing (largest shift, ms)	–	–	3.903	0.5
h/2	P09_short_pulse	overall voltage difference (RMS, mV)	0	2.451	2.451	0.5

Fault 11: m-scale_capacitance-8d53b9976a on traub2005_testseg_all



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 11 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High	Medium X	Low	–

Notes (required for any No or Unsure):

Fault 12 of 25

Fault ID	m-scale_capacitance-8f1d58bf3d
Group	random sample
Model	Traub et al. 2005 test cell with almost all channels (traub2005_testseg_all)
Label (what the edit claims to be)	scale_capacitance: multiply the membrane capacitance by a factor; factor 0.5
Severity	strong
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P04_step_2x, P05_long_step, P06_ramp, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/TestSeg_all.cell.nml

```
@@ -136,5 +136,5 @@
<spikeThresh value="0 mV"/>

- <specificCapacitance value="1.0 uF_per_cm2"/>
+ <specificCapacitance value="0.5 uF_per_cm2"/>

<initMembPotential value="-60.0 mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

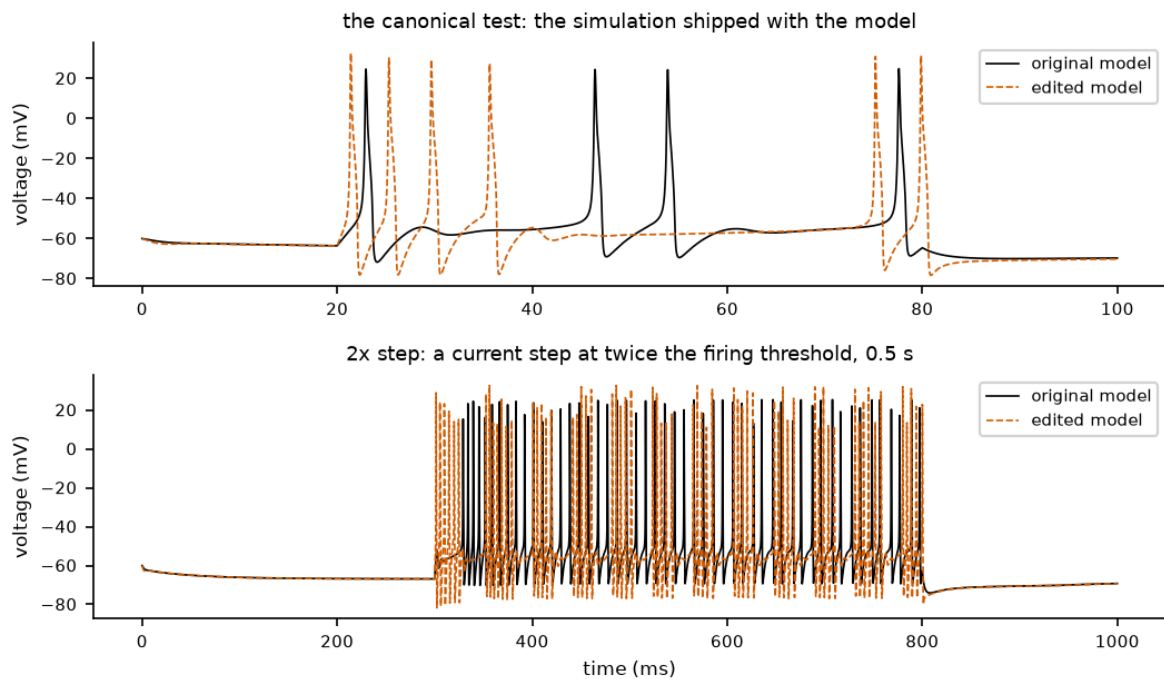
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	adaptation_index	0.5217	0.03545	0.4862	0.05217
h	P00_canonical	ahp_depth	-8.248	-14.57	6.322	0.5
h	P00_canonical	ap_amplitude	74.71	96.16	21.45	1.494
h	P00_canonical	first_spike_latency	2.95	1.4	1.55	0.5
h	P00_canonical	last_isi	23.7	4.67	19.03	0.5
h	P00_canonical	spike_count	4	6	2	0.5
h	P04_step_2x	ahp_depth	-3.064	-14.86	11.79	1.146
h	P04_step_2x	ap_amplitude	71.33	97.08	25.74	1.427
h	P04_step_2x	first_spike_latency	29.1	1.43	27.67	0.582
h	P04_step_2x	spike_count	51	63	12	3
h	P05_long_step	ahp_depth	-2.799	-15.34	12.54	1.173
h	P05_long_step	ap_amplitude	72.99	94.65	21.66	1.46
h	P05_long_step	first_spike_latency	42.89	2.04	40.85	0.8578
h	P05_long_step	last_isi	24.47	4.97	19.5	1.11
h	P05_long_step	spike_count	83	96	13	3
h	P06_ramp	spike_count	70	95	25	3

(20 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	17.29	17.29	1.009
h	P00_canonical	spike count	4	6	2	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	4.265	4.265	0.5
h	P09_short_pulse	spike timing (largest shift, ms)	–	–	0.8742	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	16.98	16.98	1.009
h/2	P00_canonical	spike count	4	6	2	0.5
h/2	P09_short_pulse	overall voltage difference (RMS, mV)	0	4.264	4.264	0.5
h/2	P09_short_pulse	spike timing (largest shift, ms)	–	–	0.8786	0.5

Fault 12: m-scale_capacitance-8f1d58bf3d on traub2005_testseg_all



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 12 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–

Notes (required for any No or Unsure):

Fault 13 of 25

Fault ID	m-scale_conductance-056f6e95e6
Group	random sample
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 2
Severity	strong
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P02_weak_step, P03_rheobase, P04_step_2x, P05_long_step, P06_ramp, P07_hyperpolarizing_step, P08_rebound, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/SomaOnly_allCml.cell.nml

```
@@ -69,5 +69,5 @@
<membraneProperties>
```

```
- <channelDensity condDensity="0.05 mS_per_cm2" id="hd_all" ionChannel="hd_vhalf1min73" ion="hd" erev="-30.0 mV"/>
+ <channelDensity condDensity="0.1 mS_per_cm2" id="hd_all" ionChannel="hd_vhalf1min73" ion="hd" erev="-30.0 mV"/>
```

```
<channelDensity condDensity="30.0 mS_per_cm2" id="kad_all" ionChannel="kad" ion="k" erev="-90.0 mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	adaptation_index	–	2.134e-13	–	inf
h	P00_canonical	ahp_depth	-11.85	-16.24	4.387	0.5927
h	P00_canonical	first_spike_latency	5.42	2.72	2.7	0.5
h	P00_canonical	last_isi	18.33	17.23	1.1	0.5
h	P00_canonical	spike_count	3	4	1	0.5
h	P02_weak_step	spike_count	0	11	11	0.5
h	P02_weak_step	steady_state_voltage	-65.02	-65.67	0.6437	0.5
h	P03_rheobase	rheobase	5.806e-4	0	5.806e-4	6.830e-5
h	P04_step_2x	ahp_depth	-14.78	-16.69	1.917	0.7388
h	P04_step_2x	first_spike_latency	16.07	16.9	0.83	0.5
h	P04_step_2x	last_isi	40.06	29.47	10.59	0.8012
h	P04_step_2x	spike_count	13	17	4	0.5
h	P05_long_step	ahp_depth	-15.07	-16.97	1.9	0.7537
h	P05_long_step	first_spike_latency	23.15	19.71	3.44	0.5
h	P05_long_step	last_isi	51.51	32.76	18.75	1.03
h	P05_long_step	spike_count	39	61	22	0.5

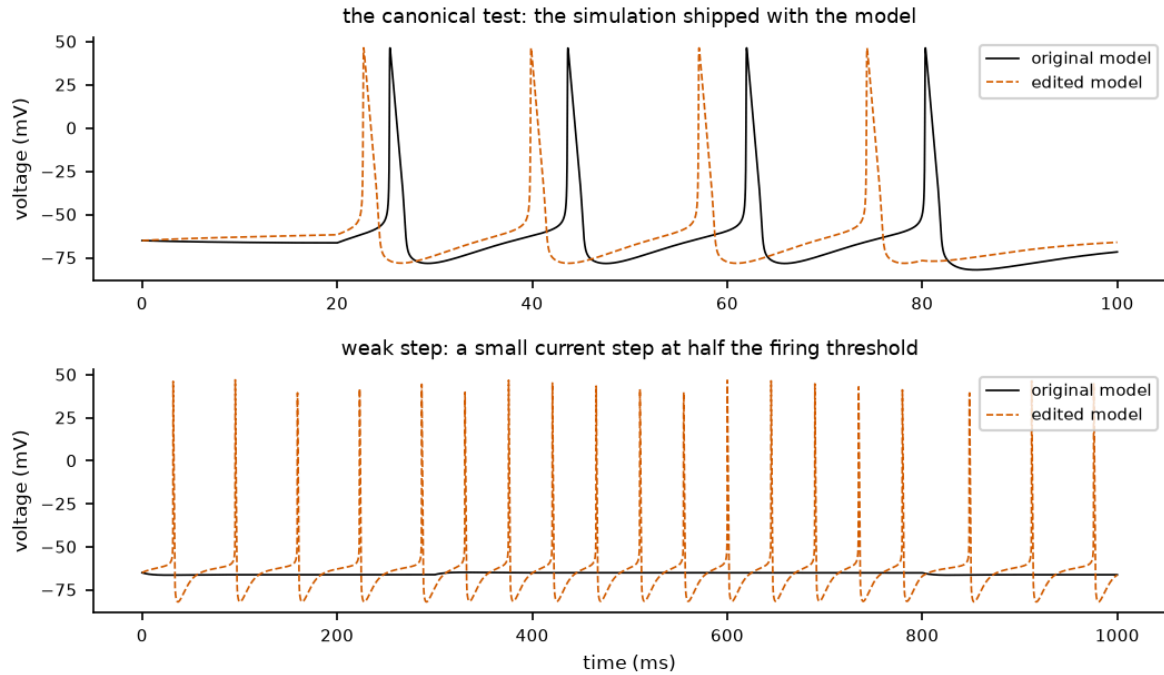
(34 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	28.66	28.66	4.056
h	P00_canonical	spike timing (largest shift, ms)	–	–	5.987	0.5
h	P02_weak_step	overall voltage difference (RMS, mV)	0	14.62	14.62	0.5
h	P02_weak_step	spike count	0	19	19	0.5
h	P04_step_2x	overall voltage difference (RMS, mV)	0	20.18	20.18	8.749
h	P04_step_2x	spike count	13	25	12	0.5
h	P05_long_step	spike count	39	69	30	0.5
h	P06_ramp	overall voltage difference (RMS, mV)	0	19.93	19.93	6.268
h	P06_ramp	spike count	16	38	22	0.5
h	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	9.833	9.833	0.5
h	P07_hyperpolarizing_step	spike count	0	8	8	0.5
h	P08_rebound	overall voltage difference (RMS, mV)	0	10.46	10.46	0.5
h	P08_rebound	spike count	0	10	10	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	13.68	13.68	1.226

(16 more rows not shown)

Fault 13: m-scale_conductance-056f6e95e6 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 13 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?			X	
Confidence: type X under one	High	Medium X	Low	—

Notes (required for any No or Unsure):

$0.05 \times 2 = 0.1$ exactly. Most feature rows are robust. However, one row lists Tolerance = inf, which cannot logically be exceeded. This is a different and more serious category of issue than a thin margin, and this is why Q3 stays Unsure regardless of how strong the other rows are regardless of how strong the other rows are.

Fault 14 of 25

Fault ID	m-scale_conductance-2956a62118
Group	random sample
Model	Blue Brain Project 2015 cortical cell (soma only) (bbp2015_soma)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 0.9
Severity	mild
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P04_step_2x

The exact change (Question 1)

File: NMC/NeuroML2/Soma_A11NML2.cell.nml

```
@@ -88,5 +88,5 @@
<channelDensity condDensity="293.0 mS_per_cm2" id="SKv3_1_all" ionChannel="SKv3_1" ion="k" erev="-85.0 mV"/>
```

```
- <channelDensityNernst condDensity="3.43 mS_per_cm2" id="Ca_LVAst_all" ionChannel="Ca_LVAst" ion="ca"/>
+ <channelDensityNernst condDensity="3.087 mS_per_cm2" id="Ca_LVAst_all" ionChannel="Ca_LVAst" ion="ca"/>
```

```
<channelDensityNernst condDensity="0.992 mS_per_cm2" id="Ca_HVA_all" ionChannel="Ca_HVA" ion="ca"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike_count	30	29	1	0.5
h/2	P04_step_2x	spike_count	30	29	1	0.5

Whole-voltage-trace differences that exceeded tolerance:

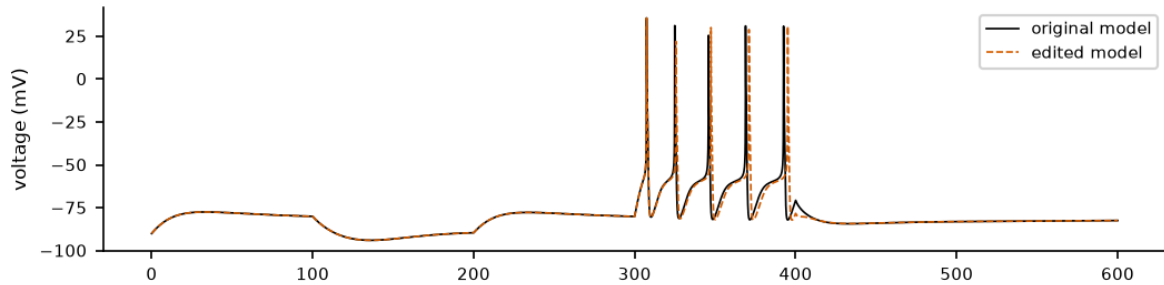
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	7.042	7.042	1.567
h	P00_canonical	spike timing (largest shift, ms)	–	–	2.613	0.5
h	P04_step_2x	spike count	30	29	1	0.5
h	P05_long_step	spike timing (largest shift, ms)	–	–	21.71	9.523
h	P06_ramp	spike timing (largest shift, ms)	–	–	3.08	2.334
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	7.045	7.045	1.567
h/2	P00_canonical	spike timing (largest shift, ms)	–	–	2.615	0.5
h/2	P04_step_2x	spike count	30	29	1	0.5
h/2	P05_long_step	spike timing (largest shift, ms)	–	–	21.77	9.523
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	3.097	2.334
h/4	P00_canonical	overall voltage difference (RMS, mV)	0	7.046	7.046	1.567

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h/4	P00_canonical	spike timing (largest shift, ms)	–	–	2.616	0.5
h/4	P04_step_2x	spike timing (largest shift, ms)	–	–	7.297	3.458
h/4	P05_long_step	spike timing (largest shift, ms)	–	–	21.81	9.523

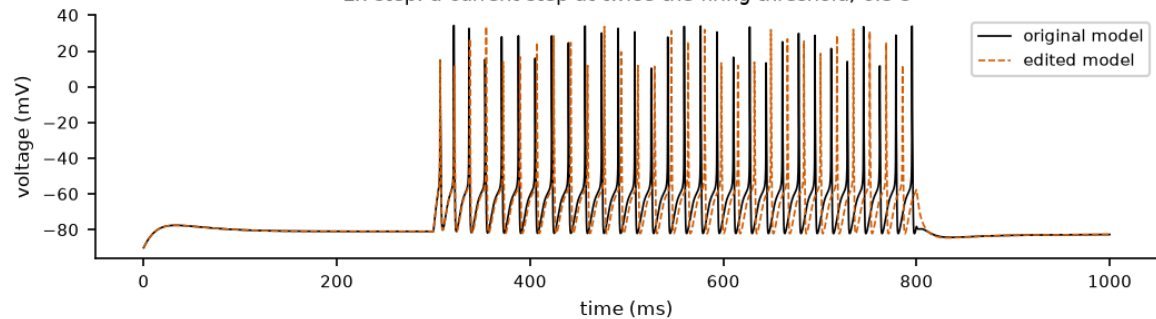
(1 more rows not shown)

Fault 14: m-scale_conductance-2956a62118 on bbp2015_soma

the canonical test: the simulation shipped with the model



2x step: a current step at twice the firing threshold, 0.5 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 14 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure):				

Fault 15 of 25

Fault ID	m-scale_conductance-c2b2371643
Group	random sample
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 0.9
Severity	mild
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P04_step_2x, P05_long_step, P06_ramp

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/SomaOnly_allCml.cell.nml

@@ -77,5 +77,5 @@

```
<channelDensity condDensity="10.0 mS_per_cm2" id="kdr_all" ionChannel="kdr" ion="k" erev="-90.0 mV"/>
```

```
- <channelDensity condDensity="25.0 mS_per_cm2" id="na3_all" ionChannel="na3" ion="na" erev="50.0 mV"/>
+ <channelDensity condDensity="22.5 mS_per_cm2" id="na3_all" ionChannel="na3" ion="na" erev="50.0 mV"/>
```

```
<channelDensity condDensity="125.0 mS_per_cm2" id="nax_all" ionChannel="nax" ion="na" erev="50.0 mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike_count	13	12	1	0.5
h	P05_long_step	first_spike_latency	23.15	23.93	0.78	0.5
h	P05_long_step	last_isi	51.51	52.63	1.12	1.03
h	P05_long_step	spike_count	39	38	1	0.5
h	P06_ramp	first_spike_latency	396.2	407	10.79	7.924
h/2	P04_step_2x	spike_count	13	12	1	0.5
h/2	P05_long_step	first_spike_latency	23.11	23.89	0.78	0.5
h/2	P05_long_step	last_isi	51.56	52.68	1.12	1.03
h/2	P05_long_step	spike_count	39	38	1	0.5
h/2	P06_ramp	first_spike_latency	396.1	406.9	10.79	7.924

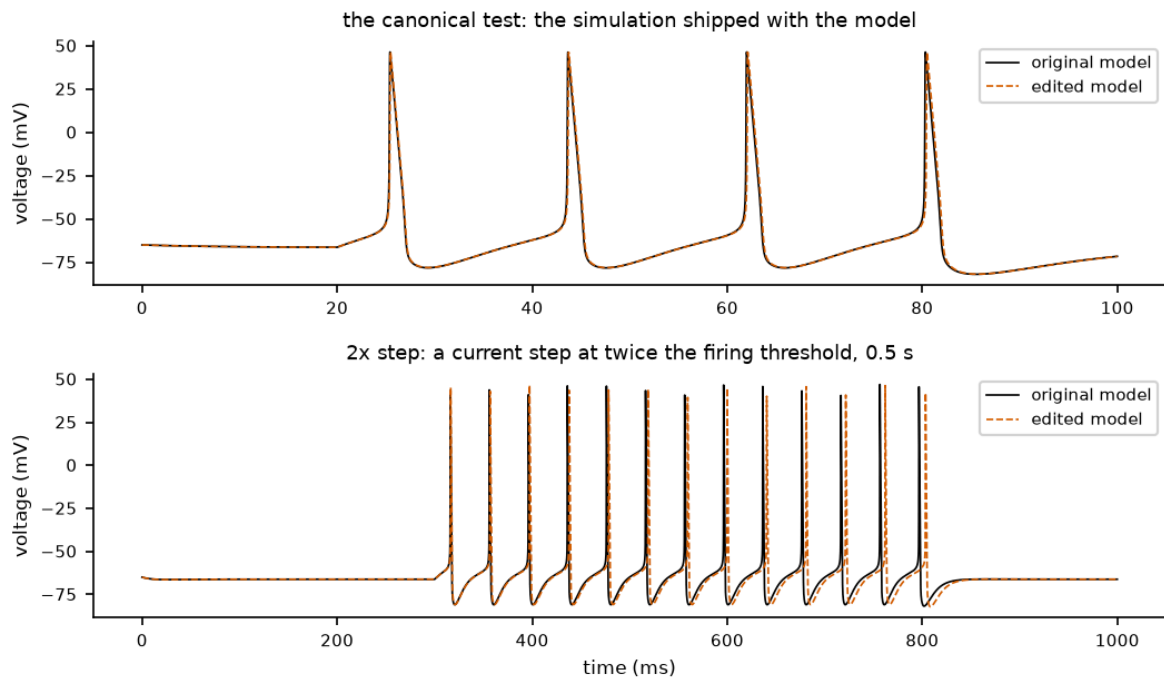
Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	4.96	4.96	4.056
h	P04_step_2x	overall voltage difference (RMS, mV)	0	15.21	15.21	8.749

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P04_step_2x	spike timing (largest shift, ms)	–	–	6.707	0.5184
h	P05_long_step	spike count	39	38	1	0.5
h	P06_ramp	overall voltage difference (RMS, mV)	0	15.69	15.69	6.268
h	P06_ramp	spike timing (largest shift, ms)	–	–	10.79	0.5781
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	4.952	4.952	4.056
h/2	P04_step_2x	overall voltage difference (RMS, mV)	0	15.16	15.16	8.749
h/2	P04_step_2x	spike timing (largest shift, ms)	–	–	6.596	0.5184
h/2	P05_long_step	spike count	39	38	1	0.5
h/2	P06_ramp	overall voltage difference (RMS, mV)	0	15.64	15.64	6.268
h/2	P06_ramp	spike timing (largest shift, ms)	–	–	10.79	0.5781
h/4	P00_canonical	overall voltage difference (RMS, mV)	0	4.95	4.95	4.056
h/4	P04_step_2x	overall voltage difference (RMS, mV)	0	15.14	15.14	8.749

(4 more rows not shown)

Fault 15: m-scale_conductance-c2b2371643 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 15 (type X)	Yes	No	Unsure	N/A
------------------------------------	-----	----	--------	-----

Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure):				

Fault 16 of 25

Fault ID	m-scale_conductance-c9ac5a125c
Group	random sample
Model	Traub et al. 2005 test cell 2 (sodium/potassium channels) (traub2005_testseg2)
Label (what the edit claims to be)	scale_conductance: multiply one ion channel's maximal conductance (how many channels there are) by a factor; factor 0.5
Severity	strong
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P02_weak_step, P03_rheobase, P04_step_2x, P05_long_step, P06_ramp, P07_hyperpolarizing_step, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/TestSeg_test2.cell.nml

@@ -69,5 +69,5 @@

<channelDensity condDensity="0.1 mS_per_cm2" id="k2_all" ionChannel="k2" ion="k" erev="-95.0 mV"/>

- <channelDensity condDensity="15.0 mS_per_cm2" id="ka_all" ionChannel="ka" ion="k" erev="-95.0 mV"/>
+ <channelDensity condDensity="7.5 mS_per_cm2" id="ka_all" ionChannel="ka" ion="k" erev="-95.0 mV"/>

<channelDensity condDensity="15.0 mS_per_cm2" id="ka_ib_all" ionChannel="ka_ib" ion="k" erev="-95.0 mV"/>

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	ahp_depth	-7.977	-6.885	1.092	0.5
h	P00_canonical	ap_amplitude	76.04	80.33	4.287	1.521
h	P00_canonical	first_spike_latency	2.91	2.33	0.58	0.5
h	P00_canonical	last_isi	7.59	6.74	0.85	0.5
h	P00_canonical	spike_count	9	10	1	0.5
h	P02_weak_step	spike_count	0	2	2	0.5
h	P02_weak_step	steady_state_voltage	-63.11	-59.61	3.499	0.5
h	P03_rheobase	rheobase	0.01711	0.007069	0.01004	3.422e-4
h	P04_step_2x	ahp_depth	-5.433	-6.819	1.386	1.145
h	P04_step_2x	first_spike_latency	53.73	21.86	31.87	1.075
h	P04_step_2x	last_isi	25.63	15.8	9.83	1.29
h	P04_step_2x	spike_count	18	32	14	0.5
h	P05_long_step	ahp_depth	-5.301	-6.698	1.398	1.153
h	P05_long_step	first_spike_latency	86.5	30.79	55.71	1.73
h	P05_long_step	last_isi	46.56	22.43	24.13	2.61
h	P05_long_step	spike_count	42	89	47	3

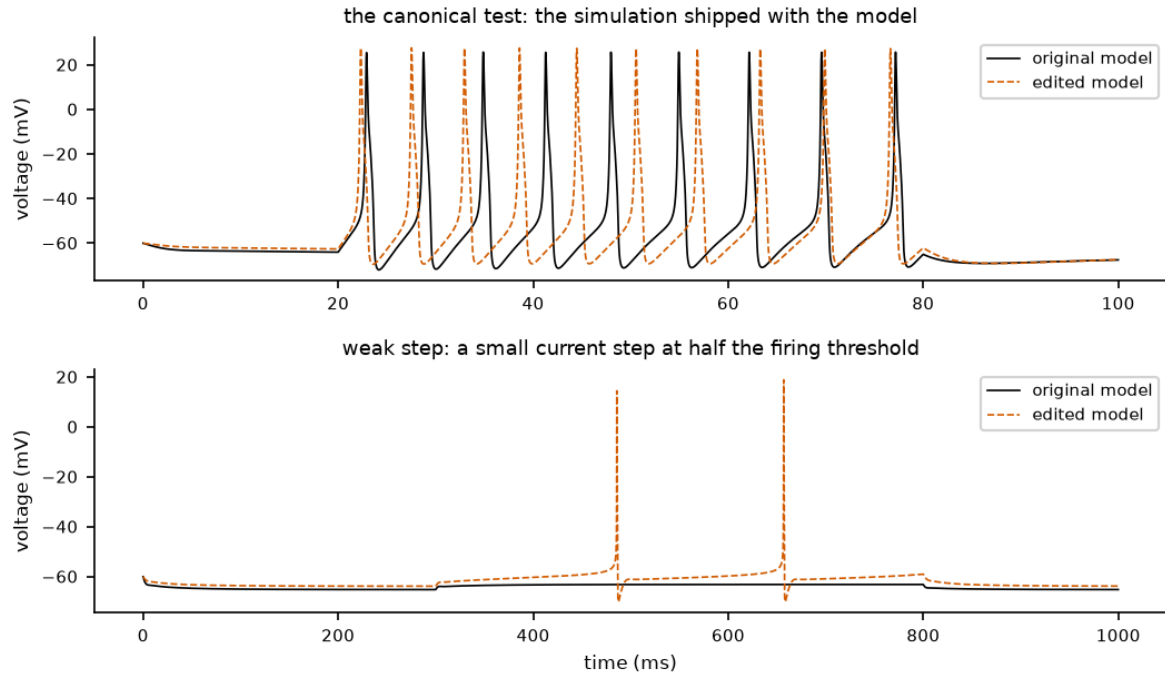
(24 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	22.32	22.32	4.39
h	P00_canonical	spike count	9	10	1	0.5
h	P02_weak_step	overall voltage difference (RMS, mV)	0	3.586	3.586	0.5
h	P02_weak_step	spike count	0	2	2	0.5
h	P04_step_2x	spike count	18	32	14	0.5
h	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	0.998	0.998	0.5
h	P08_rebound	overall voltage difference (RMS, mV)	0	0.9346	0.9346	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	1.941	1.941	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	22.33	22.33	4.39
h/2	P00_canonical	spike count	9	10	1	0.5
h/2	P02_weak_step	overall voltage difference (RMS, mV)	0	3.584	3.584	0.5
h/2	P02_weak_step	spike count	0	2	2	0.5
h/2	P04_step_2x	spike count	18	31	13	0.5
h/2	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	0.998	0.998	0.5

(2 more rows not shown)

Fault 16: m-scale_conductance-c9ac5a125c on traub2005_testseg2



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 16 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High	Medium X	Low	—
Notes (required for any No or Unsure): P00_canonical detected at h. h/2 rows truncated but trace table confirms P00_canonical flagged at h/2 too.				

Fault 17 of 25

Fault ID	m-scale_gate_slope-039828a4db
Group	random sample
Model	Blue Brain Project 2015 cortical cell (soma only) (bbp2015_soma)
Label (what the edit claims to be)	scale_gate_slope: make one channel gate's voltage dependence steeper or shallower; factor 0.8
Severity	mild
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P04_step_2x, P05_long_step, P07_hyperpolarizing_step, P09_short_pulse

The exact change (Question 1)

File: NMC/NeuroML2/Ih.channel.nml

@@ -28,6 +28,6 @@

```
<gate id="m" type="gateHHrates" instances="1">
- <forwardRate type="HHExpLinearRate" rate="0.076517per_ms" scale="-11.9mV" midpoint="-154.9mV"/>
- <reverseRate type="HHExpRate" rate="0.193per_ms" scale="33.1mV" midpoint="0mV"/>
+ <forwardRate type="HHExpLinearRate" rate="0.076517per_ms" scale="-9.52mV" midpoint="-154.9mV"/>
+ <reverseRate type="HHExpRate" rate="0.193per_ms" scale="26.48mV" midpoint="0mV"/>
</gate>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	ahp_depth	-0.5401	0.6559	1.196	0.5
h	P00_canonical	last_isi	23.59	24.1	0.51	0.5
h	P04_step_2x	ahp_depth	0.8735	3.67	2.796	0.5
h	P05_long_step	ahp_depth	0.5049	3.214	2.709	0.5
h	P05_long_step	first_spike_latency	10.13	10.9	0.77	0.5
h	P07_hyperpolarizing_step	steady_state_voltage	-97.97	-99.4	1.43	0.5
h	P09_short_pulse	ahp_depth	-0.9664	1.855	2.821	0.5
h/2	P00_canonical	ahp_depth	-0.5443	0.6516	1.196	0.5
h/2	P00_canonical	last_isi	23.58	24.1	0.52	0.5
h/2	P04_step_2x	ahp_depth	0.855	3.651	2.796	0.5
h/2	P05_long_step	ahp_depth	0.5179	3.232	2.714	0.5
h/2	P05_long_step	first_spike_latency	10.1	10.87	0.77	0.5
h/2	P07_hyperpolarizing_step	steady_state_voltage	-97.97	-99.4	1.43	0.5
h/2	P09_short_pulse	ahp_depth	-0.995	1.818	2.813	0.5

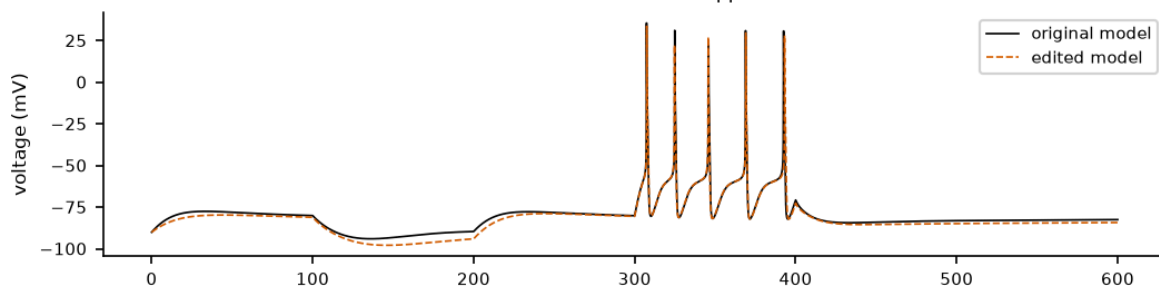
Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	3.809	3.809	1.567
h	P00_canonical	spike timing (largest shift, ms)	–	–	0.6292	0.5
h	P02_weak_step	overall voltage difference (RMS, mV)	0	1.633	1.633	0.5
h	P04_step_2x	spike timing (largest shift, ms)	–	–	4.53	3.458
h	P05_long_step	spike timing (largest shift, ms)	–	–	22.07	9.523
h	P06_ramp	spike timing (largest shift, ms)	–	–	3.322	2.334
h	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	2.063	2.063	0.5
h	P08_rebound	overall voltage difference (RMS, mV)	0	1.517	1.517	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	1.848	1.848	0.647
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	3.805	3.805	1.567
h/2	P00_canonical	spike timing (largest shift, ms)	–	–	0.6269	0.5
h/2	P02_weak_step	overall voltage difference (RMS, mV)	0	1.633	1.633	0.5
h/2	P04_step_2x	spike timing (largest shift, ms)	–	–	4.51	3.458
h/2	P05_long_step	spike timing (largest shift, ms)	–	–	22.1	9.523

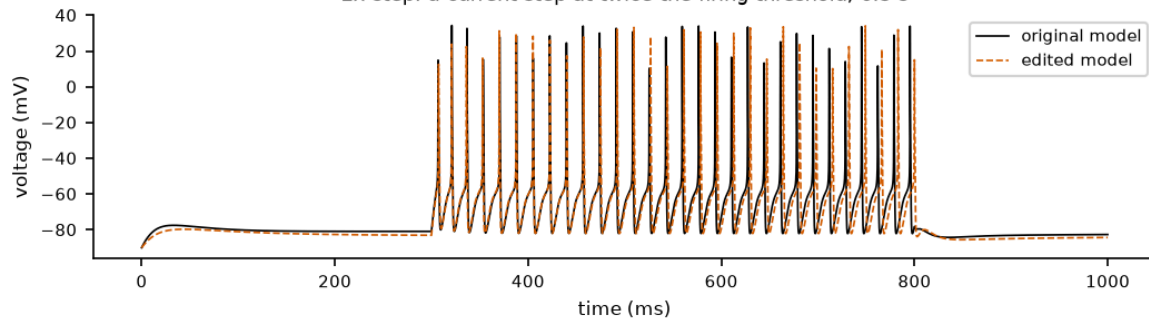
(4 more rows not shown)

Fault 17: m-scale_gate_slope-039828a4db on bbp2015_soma

the canonical test: the simulation shipped with the model



2x step: a current step at twice the firing threshold, 0.5 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 17 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure):				

Fault 18 of 25

Fault ID	m-scale_gate_time_constant-97ff619aea
Group	random sample
Model	Smith et al. 2013 layer-2/3 pyramidal cell (one compartment) (smith2013_singlecomp)
Label (what the edit claims to be)	scale_gate_time_constant: make one channel gate open and close faster or slower (scale its time constant); factor 0.5
Severity	strong
Run status	ok
Assigned class	6_silent_under_canonical: behaviour changed, but the canonical test missed it (another protocol caught it)
Detected (features) by	P03_rheobase, P06_ramp

The exact change (Question 1)

File: NeuroML2/it.channel.nml

@@ -48,4 +48,5 @@

```
<gate id="h" type="gateHHTauInf" instances="1">
+ <q10Settings type="q10Fixed" fixedQ10="0.5"/>
<timeCourse type="CaT_h_tau_tau"/>
<steadyState type="HHSigmoidVariable" rate="1" scale="-5.0mV" midpoint="-78mV"/>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

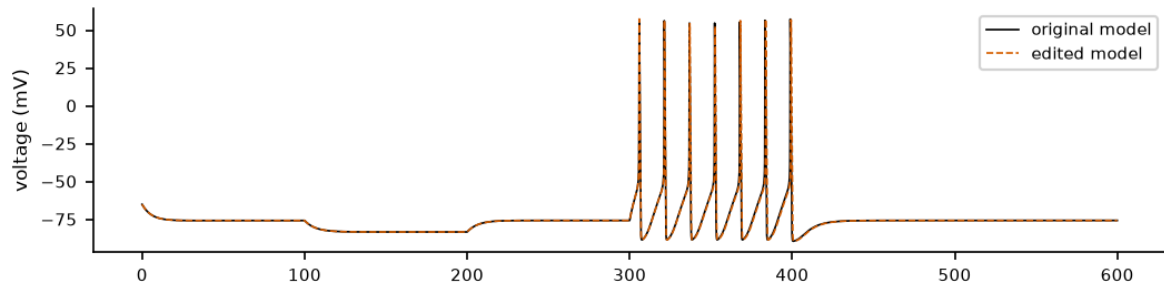
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P03_rheobase	rheobase	0.02165	0.02237	7.172e-4	4.330e-4
h	P06_ramp	first_spike_latency	403.1	389.2	13.89	8.062
h	P06_ramp	spike_count	35	36	1	0.5
h/2	P03_rheobase	rheobase	0.02165	0.02237	7.172e-4	4.330e-4
h/2	P06_ramp	first_spike_latency	403	389.1	13.88	8.062
h/2	P06_ramp	spike_count	35	36	1	0.5

Whole-voltage-trace differences that exceeded tolerance:

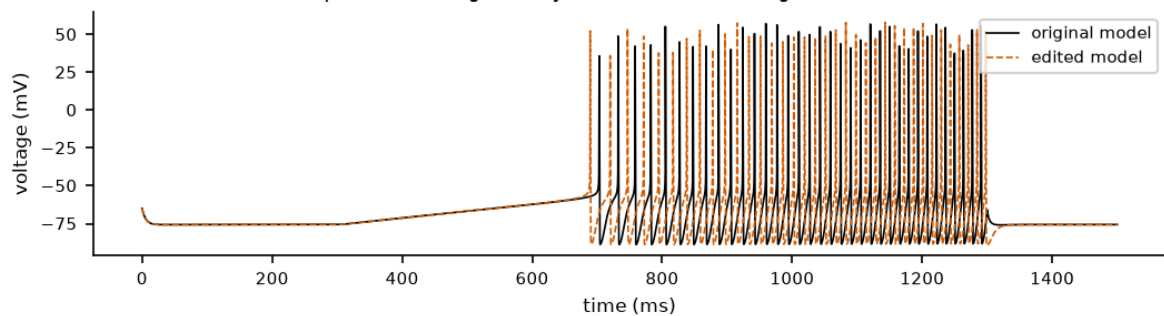
Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	6.327	6.327	3.009
h	P06_ramp	spike count	35	36	1	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	6.328	6.328	3.009
h/2	P06_ramp	spike count	35	36	1	0.5
h/4	P00_canonical	overall voltage difference (RMS, mV)	0	6.328	6.328	3.009
h/4	P06_ramp	spike count	35	36	1	0.5

Fault 18: m-scale_gate_time_constant-97ff619aea on smith2013_singlecomp

the canonical test: the simulation shipped with the model



ramp: current rising steadily from 0 to 3x the firing threshold over 1 s



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 18 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?			X	–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High	Medium X	Low	–

Notes (required for any No or Unsure):

Same concern like before in fault 4, which is that Q10 is not a direct time constant multiplier, so I am not able to confirm from the diff alone that this produces exactly a 0.5 x τ scaling as the label says.

Fault 19 of 25

Fault ID	m-shift_forward_rate_midpoint-3567e7da5d
Group	random sample
Model	Traub et al. 2005 test cell with almost all channels (traub2005_testseg_all)
Label (what the edit claims to be)	shift_forward_rate_midpoint: shift the voltage dependence of one gate's opening rate; shift 5 mV
Severity	mild
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P02_weak_step, P03_rheobase, P04_step_2x, P05_long_step, P06_ramp

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/cal.channel.nml

@@ -44,5 +44,5 @@

```
<gate id="m" type="gateHHrates" instances="2">
- <forwardRate type="HHSigmoidRate" rate="1.6per_ms" scale="13.888889mV" midpoint="5mV"/>
+ <forwardRate type="HHSigmoidRate" rate="1.6per_ms" scale="13.888889mV" midpoint="10mV"/>
<reverseRate type="HHEXPLinearRate" rate="0.1per_ms" scale="-5mV" midpoint="-8.9mV"/>
</gate>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

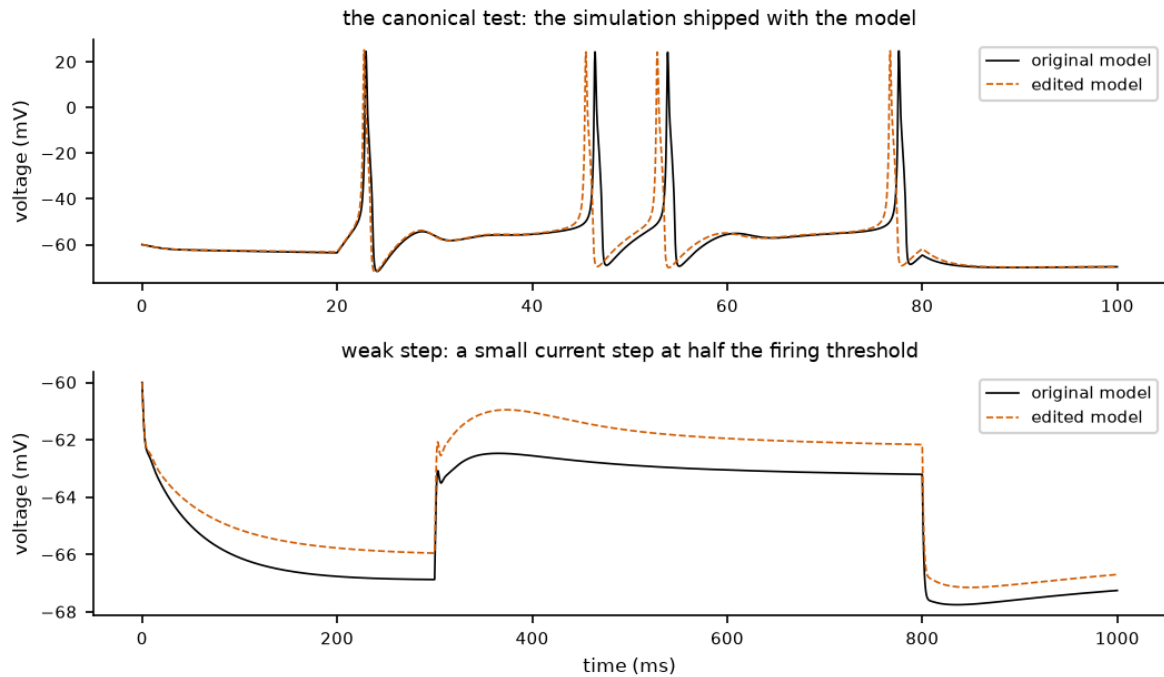
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	ahp_depth	-8.248	-8.96	0.7112	0.5
h	P02_weak_step	steady_state_voltage	-63.19	-62.15	1.043	0.5
h	P03_rheobase	rheobase	0.06144	0.04535	0.01608	0.001229
h	P04_step_2x	ahp_depth	-3.064	-9.667	6.602	1.146
h	P04_step_2x	ap_amplitude	71.33	68.96	2.372	1.427
h	P04_step_2x	first_spike_latency	29.1	3.03	26.07	0.582
h	P05_long_step	ahp_depth	-2.799	-5.169	2.37	1.173
h	P05_long_step	first_spike_latency	42.89	26.93	15.96	0.8578
h	P06_ramp	first_spike_latency	382.9	288	94.83	7.657
h/2	P00_canonical	ahp_depth	-8.214	-8.924	0.7108	0.5
h/2	P02_weak_step	steady_state_voltage	-63.19	-62.15	1.043	0.5
h/2	P03_rheobase	rheobase	0.06144	0.04535	0.01608	0.001229
h/2	P04_step_2x	ahp_depth	-2.682	-9.322	6.639	1.146
h/2	P04_step_2x	ap_amplitude	71	68.8	2.204	1.427
h/2	P04_step_2x	first_spike_latency	29.03	2.98	26.05	0.582
h/2	P05_long_step	ahp_depth	-2.408	-4.797	2.388	1.173

(2 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	12.84	12.84	1.009
h	P00_canonical	spike timing (largest shift, ms)	–	–	1.068	0.5
h	P02_weak_step	overall voltage difference (RMS, mV)	0	1.021	1.021	0.5
h	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	0.654	0.654	0.5
h	P08_rebound	overall voltage difference (RMS, mV)	0	0.6808	0.6808	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	1.064	1.064	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	12.85	12.85	1.009
h/2	P00_canonical	spike timing (largest shift, ms)	–	–	1.071	0.5
h/2	P02_weak_step	overall voltage difference (RMS, mV)	0	1.021	1.021	0.5
h/2	P07_hyperpolarizing_step	overall voltage difference (RMS, mV)	0	0.654	0.654	0.5
h/2	P08_rebound	overall voltage difference (RMS, mV)	0	0.6808	0.6808	0.5
h/2	P09_short_pulse	overall voltage difference (RMS, mV)	0	1.063	1.063	0.5

Fault 19: m-shift_forward_rate_midpoint-3567e7da5d on traub2005_testseg_all



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 19 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure):				

Fault 20 of 25

Fault ID	m-shift_gate_midpoint-0bf496636d
Group	random sample
Model	Traub et al. 2005 test cell 2 (sodium/potassium channels) (traub2005_testseg2)
Label (what the edit claims to be)	shift_gate_midpoint: shift the voltage at which one channel gate is half open; shift 10 mV
Severity	strong
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P04_step_2x, P05_long_step, P06_ramp

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/km.channel1.nml

@@ -44,6 +44,6 @@

```
<gate id="m" type="gateHHrates" instances="1">
- <forwardRate type="HHSigmoidRate" rate="0.02per_ms" scale="5mV" midpoint="-20mV"/>
- <reverseRate type="HHExpRate" rate="0.01per_ms" scale="-18mV" midpoint="-43mV"/>
+ <forwardRate type="HHSigmoidRate" rate="0.02per_ms" scale="5mV" midpoint="-10mV"/>
+ <reverseRate type="HHExpRate" rate="0.01per_ms" scale="-18mV" midpoint="-33mV"/>
</gate>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

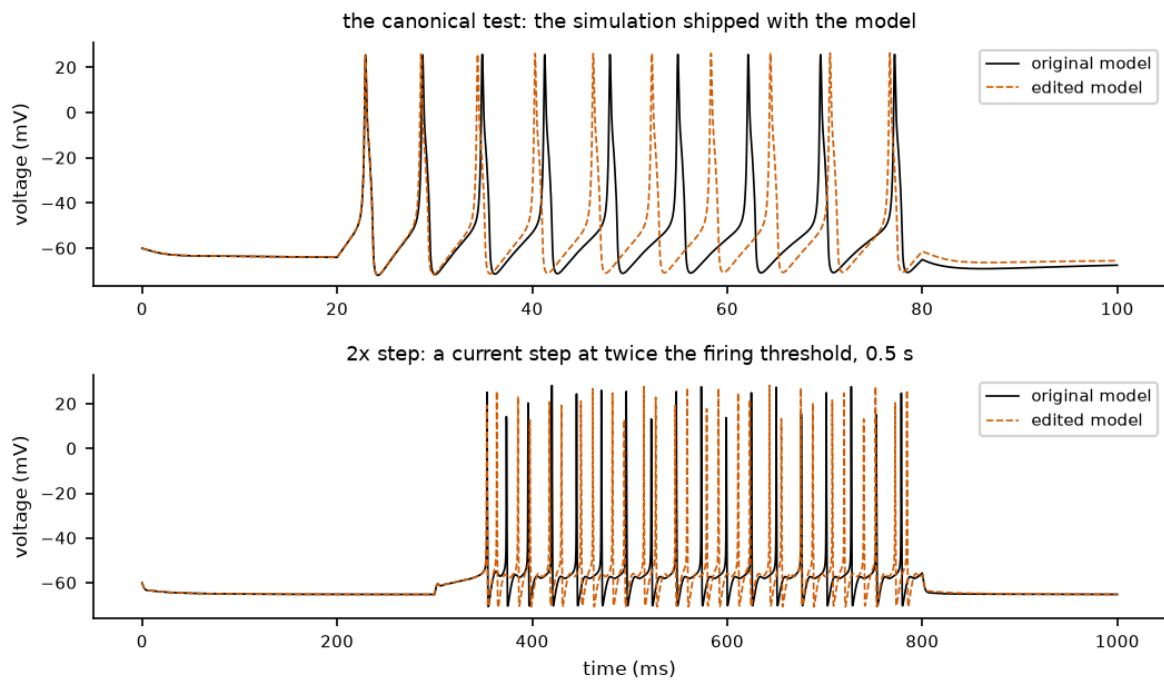
Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	adaptation_index	0.01794	0.00383	0.01411	0.01
h	P00_canonical	last_isi	7.59	6.13	1.46	0.5
h	P00_canonical	spike_count	9	10	1	0.5
h	P04_step_2x	adaptation_index	0.004861	-0.01132	0.01618	0.01
h	P04_step_2x	last_isi	25.63	12.13	13.5	1.29
h	P04_step_2x	spike_count	18	28	10	0.5
h	P05_long_step	last_isi	46.56	27.7	18.86	2.61
h	P05_long_step	spike_count	42	70	28	3
h	P06_ramp	spike_count	27	41	14	3
h/2	P00_canonical	adaptation_index	0.01802	0.004069	0.01395	0.01
h/2	P00_canonical	last_isi	7.61	6.14	1.47	0.5
h/2	P00_canonical	spike_count	9	10	1	0.5
h/2	P04_step_2x	adaptation_index	0.004952	-0.01113	0.01608	0.01
h/2	P04_step_2x	last_isi	26.06	12.19	13.87	1.29
h/2	P04_step_2x	spike_count	18	28	10	0.5
h/2	P05_long_step	last_isi	47.43	27.67	19.76	2.61

(2 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	20.1	20.1	4.39
h	P00_canonical	spike count	9	10	1	0.5
h	P04_step_2x	spike count	18	28	10	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	20.18	20.18	4.39
h/2	P00_canonical	spike count	9	10	1	0.5
h/2	P04_step_2x	spike count	18	28	10	0.5

Fault 20: m-shift_gate_midpoint-0bf496636d on traub2005_testseg2



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 20 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?	X			
Confidence: type X under one	High X	Medium	Low	—

Notes (required for any No or Unsure):

Fault 21 of 25

Fault ID	m-shift_gate_midpoint-f1b7d12414
Group	random sample
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	shift_gate_midpoint: shift the voltage at which one channel gate is half open; shift 10 mV
Severity	strong
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P02_weak_step, P03_rheobase, P04_step_2x, P05_long_step, P06_ramp, P09_short_pulse

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/nax.channel.nml

```
@@ -34,6 +34,6 @@
<gate id="m" type="gateHHratesTau" instances="3">
<q10Settings type="q10ExpTemp" q10Factor="2" experimentalTemp="24 degC"/>
- <forwardRate type="HHExpLinearRate" rate="2.88per_ms" scale="7.2mV" midpoint="-30mV"/>
- <reverseRate type="HHExpLinearRate" rate="0.8928per_ms" scale="-7.2mV" midpoint="-30mV"/>
+ <forwardRate type="HHExpLinearRate" rate="2.88per_ms" scale="7.2mV" midpoint="-20mV"/>
+ <reverseRate type="HHExpLinearRate" rate="0.8928per_ms" scale="-7.2mV" midpoint="-20mV"/>
<timeCourse type="nax_m_tau_tau"/>
</gate>
```

Evidence (Questions 2 and 3)

Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	ap_amplitude	100.1	93.24	6.9	2.003
h	P00_canonical	first_spike_latency	5.42	15.44	10.02	0.5
h	P00_canonical	last_isi	18.33	30.16	11.83	0.5
h	P00_canonical	spike_count	3	2	1	0.5
h	P02_weak_step	steady_state_voltage	-65.02	-65.99	0.9647	0.5
h	P03_rheobase	rheobase	5.806e-4	0.002732	0.002151	6.830e-5
h	P04_step_2x	adaptation_index	1.248e-5	–	–	0.01
h	P04_step_2x	ahp_depth	-14.78	–	–	0.7388
h	P04_step_2x	ap_amplitude	100.5	–	–	2.009
h	P04_step_2x	first_spike_latency	16.07	–	–	0.5
h	P04_step_2x	last_isi	40.06	–	–	0.8012
h	P04_step_2x	spike_count	13	0	13	0.5
h	P05_long_step	adaptation_index	3.158e-13	–	–	0.01
h	P05_long_step	ahp_depth	-15.07	–	–	0.7537
h	P05_long_step	ap_amplitude	100.4	–	–	2.008
h	P05_long_step	first_spike_latency	23.15	–	–	0.5

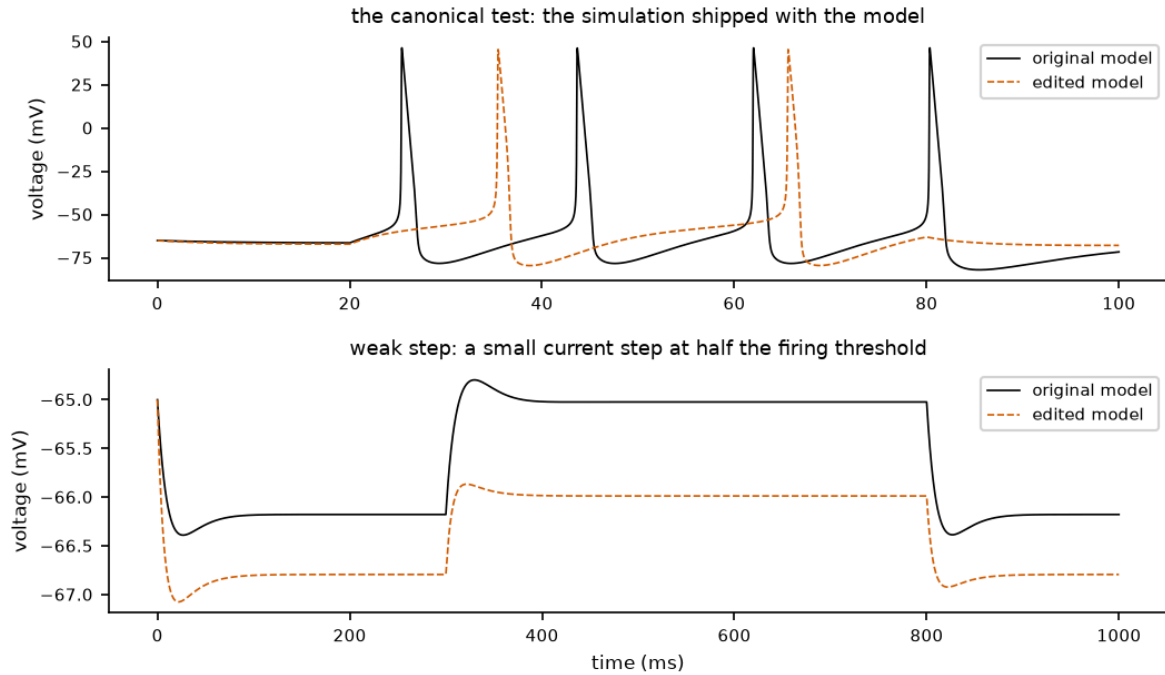
(32 more rows not shown)

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	24.75	24.75	4.056
h	P00_canonical	spike count	4	2	2	0.5
h	P02_weak_step	overall voltage difference (RMS, mV)	0	0.8133	0.8133	0.5
h	P04_step_2x	overall voltage difference (RMS, mV)	0	11.88	11.88	8.749
h	P04_step_2x	spike count	13	0	13	0.5
h	P05_long_step	spike count	39	0	39	0.5
h	P06_ramp	overall voltage difference (RMS, mV)	0	10.77	10.77	6.268
h	P06_ramp	spike count	16	0	16	0.5
h	P09_short_pulse	overall voltage difference (RMS, mV)	0	4.607	4.607	1.226
h	P09_short_pulse	spike count	1	0	1	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	24.74	24.74	4.056
h/2	P00_canonical	spike count	4	2	2	0.5
h/2	P02_weak_step	overall voltage difference (RMS, mV)	0	0.8133	0.8133	0.5
h/2	P04_step_2x	overall voltage difference (RMS, mV)	0	11.84	11.84	8.749

(6 more rows not shown)

Fault 21: m-shift_gate_midpoint-f1b7d12414 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 21 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?			X	—
Q3 Detection plausible?	X			
Confidence: type X under one	High	Medium X	Low	—

Notes (required for any No or Unsure):

Both midpoints shift -30 to -20, q10Setting context line correctly untouched. The h/2 feature rows are truncated out of the packet, and unlike some other Class 5 faults I am not able to find a direct canonical feature row. I am marking this unsure just to be sure instead of assuming its fine.

Fault 22 of 25

Fault ID	m-shift_reversal-293fd11ac3
Group	random sample
Model	Traub et al. 2005 test cell with almost all channels (traub2005_testseg_all)
Label (what the edit claims to be)	shift_reversal: shift one channel's reversal potential (the voltage its current pushes toward) by a few mV
Severity	mild
Run status	ok
Assigned class	5_non_equivalent: behaviour changed, and the canonical test caught it
Detected (features) by	P00_canonical, P05_long_step

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/TestSeg_all.cell.nml

```
@@ -123,5 +123,5 @@
<channelDensity condDensity="0.033 mS_per_cm2" id="naf_tcr_all" ionChannel="naf_tcr" ion="na" erev="50.0 mV"/>

- <channelDensity condDensity="0.05 mS_per_cm2" id="nap_all" ionChannel="nap" ion="na" erev="50.0 mV"/>
+ <channelDensity condDensity="0.05 mS_per_cm2" id="nap_all" ionChannel="nap" ion="na" erev="45.0 mV"/>

<channelDensity condDensity="15.0 mS_per_cm2" id="napf_all" ionChannel="napf" ion="na" erev="50.0 mV"/>
```

Evidence (Questions 2 and 3)

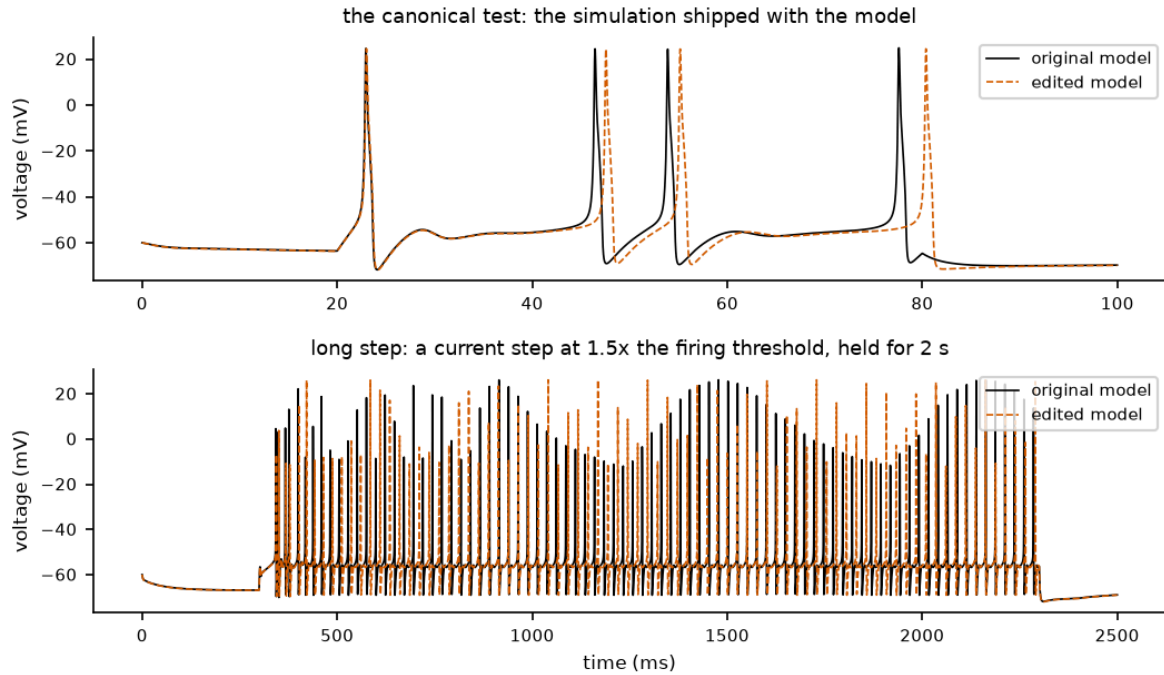
Feature differences that exceeded tolerance:

Step	Stimulus	Feature	Original	Edited	Difference	Tolerance
h	P00_canonical	adaptation_index	0.5217	–	–	0.05217
h	P00_canonical	last_isi	23.7	7.61	16.09	0.5
h	P00_canonical	spike_count	4	3	1	0.5
h	P05_long_step	first_spike_latency	42.89	43.83	0.94	0.8578
h/2	P00_canonical	adaptation_index	0.5215	–	–	0.05217
h/2	P00_canonical	last_isi	23.72	7.61	16.11	0.5
h/2	P00_canonical	spike_count	4	3	1	0.5
h/2	P05_long_step	first_spike_latency	42.82	43.76	0.94	0.8578

Whole-voltage-trace differences that exceeded tolerance:

Step	Stimulus	Measure	Original	Edited	Difference	Tolerance
h	P00_canonical	overall voltage difference (RMS, mV)	0	13.13	13.13	1.009
h	P00_canonical	spike timing (largest shift, ms)	–	–	2.773	0.5
h/2	P00_canonical	overall voltage difference (RMS, mV)	0	13.13	13.13	1.009
h/2	P00_canonical	spike timing (largest shift, ms)	–	–	2.778	0.5

Fault 22: m-shift_reversal-293fd11ac3 on traub2005_testseg_all



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 22 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			–
Q2 Assigned class plausible?	X			–
Q3 Detection plausible?			X	
Confidence: type X under one	High	Medium X	Low	–

Notes (required for any No or Unsure):

Erev 50 to 45mV, a 5mV shift, matching the few mV line comfortably. P00_canonical directly exceeds tolerance at both h and h/2. Class 5 is satisfied. I am flagging Q3 as unsure because the effect size looks disproportionate to the edit.

Fault 23 of 25

Fault ID	m-solver_config-c16c7918b3
Group	random sample
Model	Blue Brain Project 2015 cortical cell (soma only) (bbp2015_soma)
Label (what the edit claims to be)	solver_config: change the numerical integration method (a numerical stress test)
Severity	not_applicable
Run status	ok
Assigned class	4_equivalent_within_tested_domain: no reproducible change in any summary feature (behaviour looks unchanged)
Detected (features) by	nothing

The exact change (Question 1)

File: NMC/NeuroML2/LEMS_Soma_AllNML2.xml

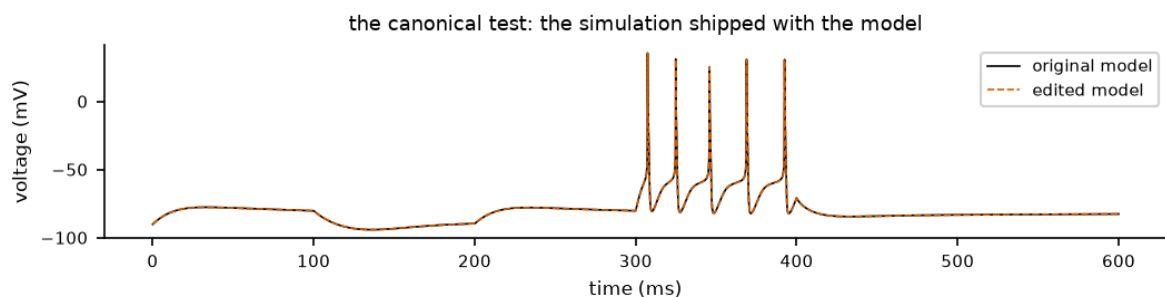
```
@@ -36,4 +36,6 @@
is used to allow validation of this file according to the LEMS schema specified above...-->
<Component type="Simulation" id="sim1" length="600.0ms" step="0.001ms" target="network_Soma_AllNML2">
+
+ <Meta for="jlems" method="eulertree"/>

<Display id="CG_TestMod_v" title="BBP Soma VOLTAGE" xmin="-60.0" xmax="660.0" ymin="-104.0" ymax="64.0"
timeScale="1ms">
```

Evidence (Questions 2 and 3)

No feature differences exceeded tolerance.

Fault 23: m-solver_config-c16c7918b3 on bbp2015_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 23 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?		X		

Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure): The header confirms that no feature differences exceeded tolerance. This is also fully consistent with class 4. There was also a Single <code><Meta for="jlems" method="eulertree" /></code> line added, nothing else touched				

Fault 24 of 25

Fault ID	m-wrong_channel-36578b9f51
Group	random sample
Model	Hay et al. 2011 layer-5 pyramidal cell (soma only) (hay2011_soma)
Label (what the edit claims to be)	wrong_channel: point a channel density at a different channel type
Severity	not_applicable
Run status	build_error
Assigned class	2_non_executable: the edited model would not run
Detected (features) by	nothing

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/Soma_AllCML.cell.nml

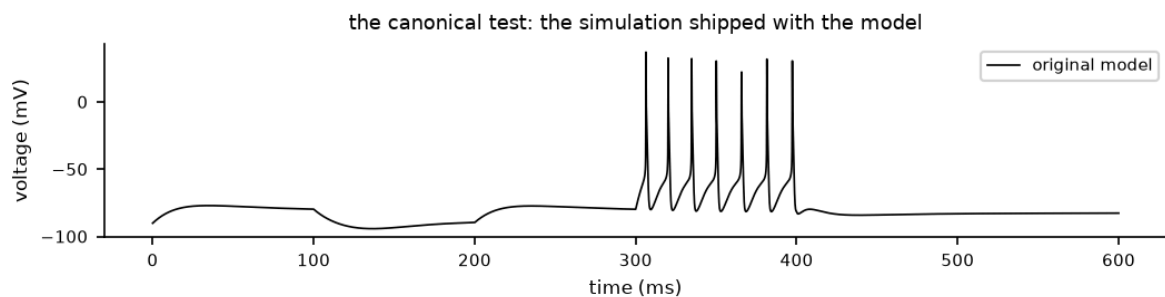
```
@@ -89,5 +89,5 @@
<!--Reversal potential for ca in Ca_LVAst will be calculated by Nernst equation from internal & external
calcium-->
```

```
- <channelDensityNernst condDensity="3.43 mS_per_cm2" id="Ca_LVAst_all" ionChannel="Ca_LVAst" ion="ca"/>
+ <channelDensityNernst condDensity="3.43 mS_per_cm2" id="Ca_LVAst_all" ionChannel="pas" ion="ca"/>
```

Evidence (Questions 2 and 3)

No feature differences exceeded tolerance.

Fault 24: m-wrong_channel-36578b9f51 on hay2011_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 24 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?		X		

Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure): ionChannel swapped from Ca_LVAst to pas while ion="ca" is left in place.				

Fault 25 of 25

Fault ID	m-wrong_compatible_component-0638e400a0
Group	random sample
Model	Migliore et al. 2005 hippocampal CA1 pyramidal cell (soma only) (migliore2005_ca1_soma)
Label (what the edit claims to be)	wrong_compatible_component: swap a channel for another channel of the same ion
Severity	not_applicable
Run status	numerically_unstable
Assigned class	3_numerically_unstable: the edited model ran but its numbers blew up
Detected (features) by	nothing

The exact change (Question 1)

File: neuroConstruct/generatedNeuroML2/SomaOnly_allCml.cell.nml

@@ -75,5 +75,5 @@

```
<channelDensity condDensity="30.0 mS_per_cm2" id="kap_all" ionChannel="kap" ion="k" erev="-90.0 mV"/>
```

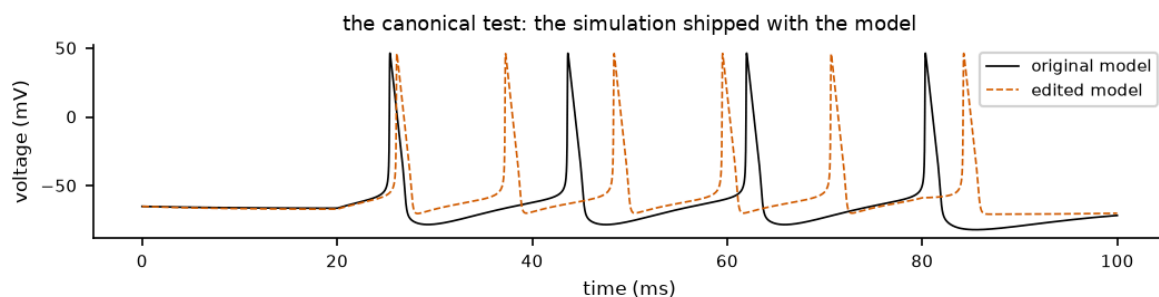
```
- <channelDensity condDensity="10.0 mS_per_cm2" id="kdr_all" ionChannel="kdr" ion="k" erev="-90.0 mV"/>
+ <channelDensity condDensity="10.0 mS_per_cm2" id="kdr_all" ionChannel="kap" ion="k" erev="-90.0 mV"/>
```

```
<channelDensity condDensity="25.0 mS_per_cm2" id="na3_all" ionChannel="na3" ion="na" erev="50.0 mV"/>
```

Evidence (Questions 2 and 3)

No feature differences exceeded tolerance.

Fault 25: m-wrong_compatible_component-0638e400a0 on migliore2005_ca1_soma



Black: original model. Orange dashed: edited model. Where the lines separate, the behaviour changed.

Your answers for fault 25 (type X)	Yes	No	Unsure	N/A
Q1 Edit matches its label?	X			—
Q2 Assigned class plausible?	X			—
Q3 Detection plausible?		X		

Confidence: type X under one	High X	Medium	Low	–
Notes (required for any No or Unsure): ionChannel swapped kdr→kap, both potassium channels with ion="k" and erev left identical				